

Does Intensity-Adjusted Volatility predict Market Risk Better than Naive Models?

A High-Frequency Duration based Volatility Forecasting Approach

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Abstract

Volatility forecasting is central to financial econometrics and risk management, yet standard GARCH-type models are designed for equally spaced returns and neglect the irregular timing of ultra-high-frequency (UHF) data. This dissertation examines whether incorporating market intensity, proxied by inter-trade durations via Autoregressive Conditional Duration (ACD) models improves volatility forecasts and risk measures relative to naïve GARCH approaches.

The empirical analysis uses BP plc tick data from April 2020, a period of extreme energy-market turbulence during the COVID-19 crisis. Durations were modelled with a Weibull-ACD(2,2), which captured strong persistence while accommodating over-dispersion beyond the exponential benchmark. The resulting conditional durations were used to construct intensity-adjusted returns, aligning volatility measures with information arrival. These adjusted and raw returns were then filtered through GARCH-family models (sGARCH, eGARCH, GJR) under Gaussian and Student- t innovations.

The results show that intensity scaling materially reduces the forecast loss. Out-of-sample evaluation with QLIKE and MSE confirmed that adjusted series enabled parsimonious Gaussian GARCH models to outperform more complex raw-return benchmarks. Diebold–Mariano tests verified that improvements were statistically significant. PIT diagnostics indicated that while Gaussian models remained under-dispersed, Student- t innovations brought calibration closer to the uniform benchmark. Value-at-Risk backtests demonstrated improved reliability at the 1% level, though clustering of exceedances persisted at 5%, pointing to remaining asymmetries.

Contents

1	Introduction	1
1.1	Purpose and Motivation	1
1.2	Market Context: Energy Volatility in April 2020	2
1.3	Problem Statement	2
1.4	High-Frequency Data and Risk Modelling	2
1.5	Research Questions	3
1.6	Contributions	3
1.7	Roadmap	4
2	Literature Review	5
2.1	Introduction to Volatility modelling	5
2.2	Extensions of GARCH: Asymmetry and Long Memory	6
2.2.1	Notation and set-up	6
2.2.2	EGARCH	6
2.2.3	GJR–GARCH	6
2.2.4	FIGARCH	7
2.3	Duration and Market Microstructure Models	7
2.3.1	Market microstructure and irregular timing	7
2.3.2	Autoregressive Conditional Duration (ACD)	8
2.3.3	Residual distributions in ACD	8
2.3.4	From intensity to volatility	8
2.3.5	Hawkes processes as an alternative	9
2.4	Research Gap	9
3	Financial Data	10
3.1	Overview of Financial Market Data	10
3.2	From Daily to Ultra-High-Frequency Data	11
3.3	Characteristics of Tick Data	12
3.4	Data Description: BP Tick Data	13
3.4.1	Data source and structure	13
3.4.2	Preprocessing and variable construction	15
3.4.3	Descriptive characteristics	15
3.4.4	De-seasonalisation of durations	17
3.4.5	Stationarity and distributional diagnostics	20
3.4.6	Data summary	22

4	Methodology	23
4.1	Methodological Pipeline	23
4.2	The ACD model layer	24
4.2.1	Residual diagnostics	26
4.3	Conditional durations and intensity adjustment	29
4.3.1	Distribution of conditional states	29
4.3.2	Exponential(1) benchmark	30
4.3.3	Weibull PIT diagnostics	30
4.3.4	Summary and transition	31
4.4	Intensity-adjusted returns	31
4.4.1	GARCH on intensity-adjusted returns	32
4.5	Forecast evaluation and backtesting	36
4.6	Robustness checks	38
5	Empirical Results	41
5.1	Variance forecasting accuracy	41
5.2	Forecast comparison: Diebold–Mariano tests	42
5.3	Density calibration: PIT analysis	43
5.4	Risk evaluation: VaR backtesting	43
5.5	Comparative synthesis: raw vs adjusted returns	44
5.6	Interpretation and link to literature	44
6	Discussion	45
6.1	Interpretation of empirical results	45
6.2	Relation to existing literature	45
6.3	Practical implications	46
6.4	Limitations and future directions	46
6.5	Summary	46
7	Conclusion	47
A	Supplementary Information: Introduction	51
A.1	Energy Market Crises (2008–2020)	51
B	Supplementary Information: Literature Review	52
B.1	Derivations of GARCH Extensions	52
B.2	Extended Catalogue of ACD Residual Distributions	53
B.3	Additional Stylised Figures	53
C	Supplementary Materials for Data Description	54
C.1	Full Daily Tick Counts and Intraday Coverage	54
C.2	Seasonality Models: Specifications and Coefficients	54
C.3	Data Cleaning and Reproducibility Notes	55
C.4	Return diagnostics	56
D	Supplementary Materials for Methodology	57
D.1	Additional robustness results	57

List of Figures

1.1	From microstructure to intensity-adjusted volatility.	4
3.1	Stylised comparison of daily vs. tick-level data. Daily sampling assumes equally spaced intervals, whereas tick data reveal irregular durations Δt between events.	12
3.2	Tick counts per trading day in April 2020. Each bar reflects the total number of quote updates between 08:00 and 16:30 London time.	15
3.3	BP mid-price trajectory on 1 April 2020, illustrating intraday volatility clustering.	16
3.4	Histogram of raw durations. The heavy right tail indicates strong over-dispersion.	17
3.5	Histogram of log-durations. The log transform reduces skewness and supports stationarity analysis.	17
3.6	Intraday seasonality in durations (before adjustment). The mean duration varies systematically over the day, with clear peaks around midday.	18
3.7	Weekday-specific diurnal patterns estimated by FFF. Each panel corresponds to a trading day, illustrating persistent seasonality captured by Fourier harmonics.	18
3.8	Intraday mean duration after FFF adjustment. The systematic daily cycle is effectively removed.	19
3.9	Stability of the FFF adjustment across trading days. The ratio of raw to adjusted durations is centred around unity, with smooth daily variation.	19
3.10	Intraday mean duration after NP (median) adjustment. Although less smooth than FFF, the non-parametric approach also reduces diurnal distortions.	20
3.11	ACF and PACF of de-seasonalised log-durations. Top: FFF-adjusted series. Bottom: NP-adjusted series. Both approaches yield near-zero autocorrelations, though weak short-lag dependence persists.	21
3.12	Goodness-of-fit diagnostics: Exponential versus Weibull. Weibull provides substantially improved fit, especially in QQ and PP plots, relative to the exponential baseline.	22
4.1	Methodological pipeline. Tick data split into (i) a price path for raw returns r_t and (ii) durations x_i for ACD-based intensity $\lambda_i = 1/\psi_i$. After aligning ψ_i to the returns grid, we construct intensity-adjusted returns $r_t^* = r_t/\sqrt{\psi_t}$. Both r_t and r_t^* feed GARCH-family models; forecasts are evaluated by QLIKE/MSE, Diebold–Mariano tests, and VaR backtests.	24
4.2	ACF and PACF of standardised residuals from the Weibull ACD(2,2). No material autocorrelation remains, suggesting that the model effectively captures duration dynamics.	27
4.3	QQ and PP diagnostics for standardised residuals from the Weibull ACD(2,2). The fit is strong in the central region, with minor departures in the upper tail.	28

4.4	Histograms of conditional durations ψ_t , intensities λ_t , and residuals ε_t from the Weibull ACD(2,2). All series are strictly positive and cluster around their theoretical means.	29
4.5	QQ-plot of residuals ε_t versus Exp(1). Departures from the 45° line in the upper tail indicate heavier tails than exponential.	30
4.6	PIT diagnostics for Weibull residuals: histogram with uniform benchmark (left), and QQ-plot against U(0,1) (right). Deviations are moderate and mostly concentrated in the tails.	31
4.7	Autocorrelation functions of absolute returns (raw and intensity-adjusted). Both exhibit long memory in volatility, justifying subsequent GARCH modelling. . .	32
4.8	Forecasted test-path $\hat{\sigma}_t$ (black) versus absolute returns $ r_t^* $ (grey). The GARCH filter is run on the test set with parameters frozen from the training fit.	34
4.9	Histogram of standardised residuals z_t (test set). While centred near zero, residuals display heavy tails, especially on the downside, a stylised fact in high-frequency finance.	35
4.10	ACF of standardised residuals z_t (left) and absolute residuals $ z_t $ (right). Absence of material autocorrelation supports the adequacy of the GARCH specification.	35
4.11	PIT histogram of standardised residuals under the sGARCH–Normal model. The distribution is centred but under-dispersed, with excess mass in the centre and tails, indicating unmodelled tail risk.	37
4.12	VaR backtest on the test set (intensity-adjusted returns). Dashed: 5% VaR; solid: 1% VaR. Filled points mark exceedances.	37
4.13	PIT histogram under sGARCH–Student- t . The distribution is closer to uniform than the Gaussian case, but residual under-dispersion persists.	39
4.14	VaR backtest on the test set (sGARCH–Student- t). Dashed: 5% VaR; solid: 1% VaR. Filled points mark exceedances.	39
B.1	Impulse responses illustrating leverage (solid vs. dashed) and long-memory decay (dotted).	53
C.1	Intraday coverage and tick density by day (full month).	54
C.2	Cullen–Frey skewness–kurtosis map for log-durations.	55
C.3	Histograms and QQ plots of raw and intensity-adjusted returns against the Normal distribution.	56

List of Tables

2.1	Milestones in the evolution of volatility modelling.	6
2.2	Asymmetric and long-memory extensions: core forms and notes.	7
2.3	Calendar-time GARCH vs. event-time duration frameworks.	8
2.4	Residual distributions for ACD: properties and use cases.	8
2.5	ACD vs. Hawkes for event-time dynamics.	9
3.1	Types of financial market data across different frequencies.	11
3.2	Key characteristics of tick-by-tick data and their econometric implications.	13
3.3	Snippet of BP tick data (1 April 2020).	14
3.4	Tick counts per trading day in April 2020.	14
3.5	Daily duration summary for April 2020. Reported values are in seconds.	17
3.6	Variance of log-durations before and after de-seasonalisation.	20
4.1	Model selection for ACD specifications under alternative innovation distributions. The Weibull ACD(2,2) achieves the lowest AIC among admissible models, justifying its use as the baseline specification for subsequent intensity construction.	26
4.2	Descriptive statistics of raw and intensity-adjusted returns.	32
4.3	Out-of-sample performance on the <i>scaled</i> returns r_t^* (test set). Lower QLIKE is better.	33
4.4	VaR backtest results on the test set. Over-violations at 1% and conditional dependence of exceedances suggest that heavier-tailed or skewed innovation distributions may be required.	38
4.5	VaR backtest results (sGARCH–Student- t).	40
5.1	Out-of-sample variance forecast evaluation. Lower values indicate better performance.	42
5.2	Pairwise DM tests for QLIKE. Entry shows <i>DM stat</i> (<i>p-value</i>). Row model outperforms column if $DM > 0$ and $p < 0.05$	42
5.3	DM tests for sGARCH–Normal against alternative models (QLIKE and MSE). Bold <i>p-values</i> indicate significance at the 5% level.	43
A.1	Major episodes of oil and energy market volatility (2008–2020).	51
D.1	Full robustness ranking (test set).	57
D.2	Split-sample forecast evaluation (sGARCH–Student- t).	57

Chapter 1

Introduction

1.1 Purpose and Motivation

Volatility is a central object in modern finance. It enters option pricing and hedging, anchors the measurement of market risk, and influences portfolio construction and capital allocation. In particular, risk measures such as Value-at-Risk (VaR) and Expected Shortfall (ES) are highly sensitive to volatility inputs; when volatility is underestimated during turbulent periods, the resulting risk numbers can be materially miscalibrated, with potentially large consequences for trading limits, margining, and capital adequacy.

A clear illustration arises in April 2020, when global energy markets were hit by the joint shock of a pandemic-driven collapse in demand and severe disruption to supply coordination. During this episode, market activity alternated between frenetic bursts of trading and abrupt lulls in liquidity. BP plc—one of the largest constituents of the FTSE 100—exhibited pronounced fluctuations at the tick level, making this period a stringent and practically relevant testbed for volatility and risk forecasting.

The rise of electronic trading has also transformed the available data: ultra-high-frequency (UHF) records capture trades and quotes arriving at irregular, unevenly spaced times. Classical GARCH-type models are typically developed for regularly spaced return series; when applied mechanically to UHF returns, they do not explicitly incorporate the irregular timing of events or the clustering of market activity. Yet market behaviour is inherently time-varying: intense trading spells, quiet intervals, and abrupt information arrivals are all reflected not only in price changes, but also in the *rate* at which trades occur.

This motivates an alternative lens that views volatility through the rhythm of market activity itself. By modelling the durations between market events, it is possible to extract a time-varying measure of market intensity and use it to form *intensity-adjusted* (time-deformed) returns before applying volatility filters. Autoregressive Conditional Duration (ACD) models (Engle & Russell, 1998) provide a natural framework for this task by modelling conditional expected durations and, by implication, a conditional intensity process.

This dissertation therefore asks whether intensity-adjusted volatility—constructed by com-

binning ACD-based intensity extraction with GARCH-family volatility filters—delivers more accurate forecasts and sharper risk assessment than conventional GARCH applied directly to raw returns.

1.2 Market Context: Energy Volatility in April 2020

The empirical setting is BP plc tick data in April 2020, a month of exceptional turmoil in energy markets: a pandemic-driven demand collapse, the OPEC+ breakdown, and the historically negative WTI settlement on 20 April (Energy Information Administration, 2020). This period featured extreme price swings alongside sharp changes in trading intensity, providing a demanding environment in which to evaluate whether incorporating event-time information improves volatility and tail-risk forecasts.¹

1.3 Problem Statement

Standard calendar-time volatility models applied to UHF returns typically ignore irregular event timing. When trades bunch together, volatility often rises; when trading becomes sparse, volatility dynamics may differ. Ignoring intensity can therefore bias or destabilise forecasts precisely when accuracy is most critical. ACD models (Engle & Russell, 1998) estimate conditional expected durations and hence provide an econometric proxy for market intensity. However, there remains limited empirical consensus on whether *scaling* returns by ACD-implied durations leads to measurable improvements in volatility forecasting and downstream risk performance (e.g., VaR calibration), relative to naïve baselines that model raw returns on an unchanged calendar clock.

Core research question. *Do volatility and risk forecasts improve when GARCH-type models are applied to intensity-adjusted returns—constructed using ACD-implied conditional durations—relative to naïve GARCH applied to raw returns?*

1.4 High-Frequency Data and Risk Modelling

Ultra-high-frequency data capture markets at their most granular level, recording every trade and quote update. Such data reveal empirical regularities that can be muted in daily aggregates, including heavy tails, volatility bursts, and frictions arising from market microstructure. Crucially, the timing of trades itself conveys information: periods of intense quoting and transaction activity often coincide with heightened volatility, wider spreads, and changing liquidity conditions.

This suggests a natural bridge between microstructure and risk modelling. By explicitly modelling inter-trade durations using the Autoregressive Conditional Duration (ACD) frame-

¹A comparative summary of major energy crises (2008–2020) is provided in Appendix A.1.

work (Engle & Russell, 1998), one can link the rhythm of market activity to conditional volatility, either by time-deforming returns (to better align innovations with information arrival) or by using intensity as a volatility driver. In the context of April 2020—when BP tick data exhibited both bursts of trading and extended pauses—such intensity-aware modelling provides a principled alternative to applying conventional GARCH filters directly to raw UHF returns.

1.5 Research Questions

The dissertation addresses three linked questions, moving from modelling to evaluation:

RQ1 Duration modelling. Does an ACD specification (estimated on BP tick data in April 2020) provide a reliable and informative stand-in for market intensity?

RQ2 Forecasting layer. Does scaling returns using the ACD-implied conditional expected duration improve out-of-sample volatility forecasts from GARCH-family models, relative to naïve GARCH applied to raw returns?

RQ3 Risk relevance. Do any improvements translate into better distributional calibration (PIT) and Value-at-Risk performance? Are differences statistically significant (Diebold–Mariano tests)?

1.6 Contributions

C1 ACD at scale in crisis conditions. Estimation and diagnostics of a Weibull ACD(2,2) on more than one million durations for BP in April 2020, documenting persistence and residual properties.

C2 Intensity-adjusted returns. A reproducible pipeline that normalises transaction returns by the square root of the ACD-implied conditional expected duration (i.e., time-deformation aligned with information arrival), providing volatility filters with intensity-adjusted inputs.

C3 Head-to-head evaluation. A systematic comparison across sGARCH, eGARCH, and GJR with Normal vs. Student- t innovations, estimated on raw vs. adjusted returns. Loss functions such as QLIKE and MSE are *defined once in* Chapter 4 and referred to in Chapter 5.

C4 Economic validation. Density calibration (PIT), Value-at-Risk backtests, and Diebold–Mariano forecast comparison tests.

C5 Evidence from a stress episode. Empirical results showing whether intensity scaling materially improves variance forecasts and tail calibration during an extreme energy-market regime.

1.7 Roadmap

Chapter 2 reviews volatility, duration, and market microstructure literatures. Chapter 3 details the BP dataset and preprocessing pipeline. Chapter 4 presents the modelling framework: ACD estimation, construction of intensity-adjusted returns, GARCH filters, and formal definitions of evaluation metrics (losses, PIT, VaR, DM test). Chapter 5 reports empirical findings, with a synthesis comparing raw vs. adjusted returns. Chapter 6 interprets findings in relation to literature and practice. Chapter 7 concludes.

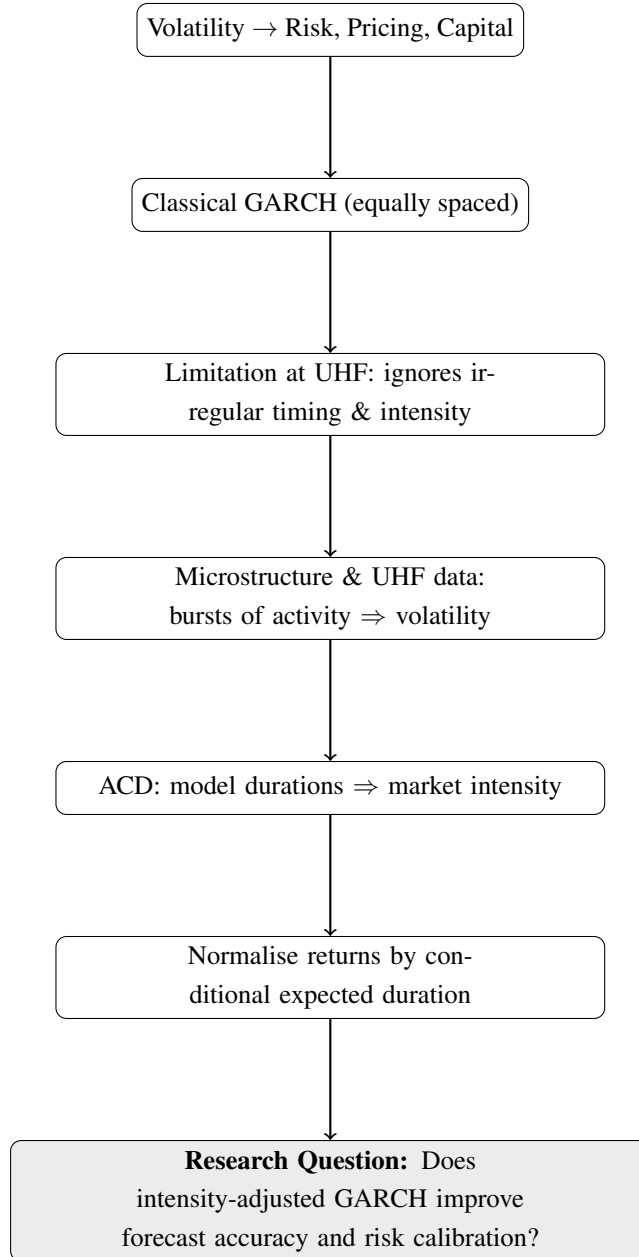


Figure 1.1: From microstructure to intensity-adjusted volatility.

Chapter 2

Literature Review

This chapter reviews the foundational literature on volatility modelling, from classical GARCH frameworks to duration-based approaches. It explains how ACD models extend standard processes to account for irregular event timing, and how these ideas support the integration of trade intensity into volatility forecasts.

2.1 Introduction to Volatility modelling

Volatility the conditional variance of asset returns is fundamental to risk management, asset pricing, and regulatory capital allocation (Engle, 1982; Bollerslev, 1986; Basel Committee on Banking Supervision, 2011). Practical use cases (portfolio choice, option valuation, VaR/ES) require models that capture salient return regularities while remaining tractable for estimation and forecasting.

The econometric treatment of time-varying volatility begins with Engle (1982), whose ARCH model lets conditional variance depend on past squared innovations, thereby reproducing volatility clustering. Bollerslev (1986) generalise to GARCH by adding lags of the conditional variance, producing a parsimonious and persistent volatility process. These innovations established the workhorse class for daily/weekly applications and remain central to regulatory-facing risk systems.

A large body of evidence shows that (G)ARCH-type models outperform naive historical-volatility rules in out-of-sample forecasting (e.g. Francq & Zakoïan, 2019). They enjoy likelihood-based estimation, closed-form filtering, and straightforward multi-step prediction. However, a key assumption is regular sampling in calendar time. In ultra-high-frequency (UHF) settings, trades and quotes occur at irregular times, and the *rate of events* (market intensity) itself co-moves with volatility (Engle & Russell, 1998; Gould et al., 2013). This motivates models that incorporate durations and point-process dynamics. Formal definitions used later (loss functions, PIT, VaR, DM) are *introduced once* in Chapter 4 and referred to thereafter, avoiding repetition across chapters (cf. Chapter 5).

Table 2.1: Milestones in the evolution of volatility modelling.

Model	Reference	Contribution
ARCH	Engle (1982)	Conditional heteroskedasticity via past squared shocks.
GARCH	Bollerslev (1986)	Adds lagged variance; parsimonious persistence.
EGARCH	Nelson (1991)	Log-variance dynamics; leverage effects.
GJR–GARCH	Glosten et al. (1993)	Threshold asymmetry for “bad news”.
FIGARCH	Baillie et al. (1996)	Fractional differencing; long memory in volatility.

2.2 Extensions of GARCH: Asymmetry and Long Memory

Two empirical features often strain the symmetric GARCH(1,1): (i) leverage (asymmetric reaction to negative vs. positive shocks) and (ii) slow, hyperbolic decay in volatility autocorrelation (long memory). We briefly review three standard extensions; their use in our empirical pipeline (Chapter 4) is to stress-test whether intensity scaling helps *across* variance specifications.

2.2.1 Notation and set-up

Let y_t be (demeaned/mean-modelled) returns with innovation $\epsilon_t = y_t - \mu_t$, variance $\sigma_t^2 = \text{Var}(\epsilon_t \mid \mathcal{F}_{t-1})$, and standardized residual $z_t = \epsilon_t / \sigma_t$. Unless stated, z_t may follow Gaussian or heavy-tailed laws (e.g. Student- t).

2.2.2 EGARCH

Model.

$$\log \sigma_t^2 = \omega + \beta \log \sigma_{t-1}^2 + \alpha (|z_{t-1}| - \mathbb{E}|z_{t-1}|) + \gamma z_{t-1}. \quad (2.1)$$

Features. Positivity is automatic (log-variance). γ governs leverage (negative shocks $\Rightarrow$ larger impact when $\gamma < 0$); β captures persistence.

2.2.3 GJR–GARCH

Model.

$$\sigma_t^2 = \omega + \alpha \epsilon_{t-1}^2 + \gamma \epsilon_{t-1}^2 \mathbb{I}\{\epsilon_{t-1} < 0\} + \beta \sigma_{t-1}^2. \quad (2.2)$$

Features. Threshold asymmetry via γ . A common diagnostic heuristic is $\alpha + \beta + \gamma/2 < 1$ for covariance-stationary-like behaviour (distribution-dependent in exact form).

2.2.4 FIGARCH

Model (FIGARCH(1,d,1)).

$$(1 - \beta L) \sigma_t^2 = \omega + [1 - \beta L - \phi(1 - L)^d] \epsilon_t^2, \quad d \in [0, 1). \quad (2.3)$$

Features. d controls long memory; $d = 0$ collapses to GARCH. Useful in crisis windows with near-integrated persistence.

Table 2.2: Asymmetric and long-memory extensions: core forms and notes.

Model	Core variance equation	Notes
EGARCH(1,1)	$\log \sigma_t^2 = \omega + \beta \log \sigma_{t-1}^2 + \alpha(z_{t-1} - \mathbb{E} z_{t-1}) + \gamma z_{t-1}$	Leverage via γ ; positivity automatic.
GJR-GARCH(1,1)	$\sigma_t^2 = \omega + \alpha \epsilon_{t-1}^2 + \gamma \epsilon_{t-1}^2 \mathbb{I}\{\epsilon_{t-1} < 0\} + \beta \sigma_{t-1}^2$	Threshold asymmetry; heuristic $\alpha + \beta + \gamma/2 < 1$.
FIGARCH(1,d,1)	$(1 - \beta L) \sigma_t^2 = \omega + [1 - \beta L - \phi(1 - L)^d] \epsilon_t^2$	Hyperbolic decay; d governs memory.

Distributional choice. Tail-sensitive tasks (VaR/ES) benefit from Student- t /skew- t innovations; this choice is orthogonal to variance dynamics but material in backtesting (Chapter 5).

Link to intensity adjustment. These models operate in calendar time. In our pipeline (Chapter 4), we also apply them to *intensity-adjusted* returns $r_t^* = r_t / \sqrt{\psi_t}$, where ψ_t comes from an ACD layer. This enables a clean comparison of naive vs. adjusted forecasts across variance specifications.

2.3 Duration and Market Microstructure Models

Calendar-time GARCH assumes regular spacing, whereas UHF markets exhibit irregular event times. The *timing* of trades/quotes transmits information about market activity and volatility. Duration models explicitly characterise inter-event times and provide a bridge from microstructure to volatility forecasting (used in Chapters 4–5).

2.3.1 Market microstructure and irregular timing

Let T_i be the timestamp of the i -th market event (trade or quote) and

$$x_i = T_i - T_{i-1} \quad (2.4)$$

the duration. Short x_i indicates high activity and typically coincides with volatile conditions.

Table 2.3: Calendar-time GARCH vs. event-time duration frameworks.

	Calendar-time GARCH	Event-time duration
Time index	Regular (minutes, days)	Irregular (event arrivals)
Volatility drivers	Past shocks and variance	Inter-event times (intensity)
Pros	Tractable; widely adopted	Captures timing, valuable at UHF
Cons	Ignores irregular spacing	Estimation/diagnostics more involved

2.3.2 Autoregressive Conditional Duration (ACD)

Engle & Russell (1998) model the conditional expected duration

$$\psi_i = \omega + \alpha x_{i-1} + \beta \psi_{i-1}, \quad x_i = \psi_i \varepsilon_i, \quad \mathbb{E}[\varepsilon_i] = 1, \quad (2.5)$$

with $\varepsilon_i > 0$ drawn from a parametric family. Here, ψ_i captures predictable activity (intensity proxy), while ε_i is residual noise. The *intensity* is $\lambda_i = 1/\psi_i$.

2.3.3 Residual distributions in ACD

Duration data are positive, skewed, and often heavy-tailed; the choice of F for ε_i is critical. Common options include Exponential (benchmark), Weibull (shape-driven dispersion), and more flexible Generalised Gamma / Generalised F / Burr families. Flexibility improves fit in stress regimes but can raise numerical demands.

Table 2.4: Residual distributions for ACD: properties and use cases.

Distribution	Key properties	Notes / use cases
Exponential	Memoryless; fixed dispersion	Benchmark; often too rigid in crises
Weibull	Shape controls over-/under-dispersion	Popular, tractable; good general workhorse
Gen. Gamma	Two shape parameters; heavy tails	Stress periods; higher computational cost
Gen. F	Very flexible skew/tails (4 params)	Extreme clustering; potential instability
Burr XII	Heavy-tailed; flexible hazard	Crisis data; sometimes numerically delicate

2.3.4 From intensity to volatility

Since $\lambda_i = 1/\psi_i$ proxies activity, scaling returns by $\sqrt{\psi_t}$ yields $r_t^* = r_t/\sqrt{\psi_t}$, which partially removes heteroskedasticity due to irregular timing. In Chapter 4 we align ψ_t with returns and compare GARCH-family filters on r_t vs. r_t^* ; results are reported in Chapter 5.

2.3.5 Hawkes processes as an alternative

Hawkes models specify the conditional intensity directly via self-excitation,

$$\lambda(t) = \mu + \int_0^t g(t-s) dN(s),$$

typically with $g(u) = \alpha e^{-\beta u}$ (Bacry et al., 2015). They naturally capture clustering/contagion but are computationally heavier to embed into return-volatility filters. For our goal, a head-to-head comparison of naive vs. intensity-adjusted GARCH forecasts, ACD offers a more direct and estimable route; see Chapter 4 for the chosen pipeline.

Table 2.5: ACD vs. Hawkes for event-time dynamics.

	ACD	Hawkes
Core object	Expected duration ψ_i	Intensity $\lambda(t)$
Dynamics	ARMA-like on x_i	Self-exciting point process
Pros	Directly yields ψ_t for scaling	Rich clustering/contagion mechanism
Cons	Sensitive to F misspecification	Heavier estimation; simulation often needed
Fit for this study	Tractable for GARCH coupling	Beyond scope for integrated VaR backtests

2.4 Research Gap

Daily/intraday GARCH models ignore irregular timing; duration models capture it but are often studied in isolation from return volatility. Evidence on whether *intensity-adjusted* returns improve *out-of-sample* volatility forecasts (and *risk* metrics) is limited, especially in stress regimes. This dissertation fills that gap by (i) estimating a Weibull-ACD on BP tick data during April 2020 (extreme energy-market turbulence), (ii) constructing $r_t^* = r_t / \sqrt{\psi_t}$, and (iii) benchmarking GARCH-family forecasts on r_t vs. r_t^* using the evaluation toolkit defined in Chapter 4 and applied in Chapter 5.

Chapter 3

Financial Data

3.1 Overview of Financial Market Data

Building on Chapter 2, financial markets generate vast amounts of data across multiple temporal resolutions, ranging from annual accounting statements to ultra-high-frequency (UHF) tick-by-tick records. The choice of frequency is not merely a matter of granularity but directly shapes the econometric tools available for modelling and the type of risk that can be addressed. At coarser frequencies (e.g. monthly or daily), data provide a smoothed representation of long-run trends and portfolio risk, whereas finer resolutions (e.g. intraday or tick) reveal the microstructure dynamics of trading and order flow.

For volatility modelling, data frequency is particularly critical: while daily or weekly returns capture broad clustering, UHF data expose how volatility is linked to the timing and intensity of transactions. This study focuses on tick-level data, as it provides the highest resolution available for investigating the interaction between event arrivals (durations) and return dynamics. Table 3.1 summarises the main characteristics of financial market data across different levels of temporal aggregation.

Table 3.1: Types of financial market data across different frequencies.

Frequency	Sampling scheme	Key features	Typical applications
Annual/Quarterly	Fixed calendar periods	Long-run fundamentals; macro drivers	Asset pricing; accounting valuation
Daily	Close or regular intraday grid	Volatility clustering; ignores intraday activity	Risk management; portfolio allocation; long-horizon volatility
Intraday (5-min)	Regular grid in trading hours	Balance of detail/tractability; partial intraday volatility	Realised volatility; intraday risk
Tick-by-tick	Every trade/quote	Order flow, microstructure; irregular spacing	Microstructure studies; HFT analysis
Durations	Time between events	Direct measure of market intensity	Event-time econometrics; ACD

3.2 From Daily to Ultra-High-Frequency Data

The shift from daily or intraday returns to UHF data has transformed the empirical study of financial markets. Daily closing prices provide a convenient and widely used representation of market activity, but they compress thousands of transactions into a single observation and therefore discard much of the information generated during the trading day. Intraday data sampled at fixed intervals (e.g. 5-minute returns) partially address this by capturing some volatility dynamics, yet they still treat markets as if information flows at regular calendar-time intervals.

UHF data, on the other hand, *capture* each trade and quote update, enabling researchers to examine directly the timing and volume of market activity. The main characteristic of such data is their *irregular spacing*: trades arrive randomly, with lengthy pauses during quiet trading and spikes of activity during news or crisis periods. Since traditional econometric models usually assume equidistant time series, these irregularities present particular difficulties. At the same time, they open opportunities for modelling volatility as a joint function of both returns and the rate at which information arrives.

Figure 3.1 illustrates the contrast between daily sampling, which assumes evenly spaced observations, and tick-level sampling, which reveals the uneven durations between events. This distinction motivates the use of duration-based models such as the Autoregressive Conditional Duration (ACD) framework, which explicitly incorporates event timing into the analysis of volatility.

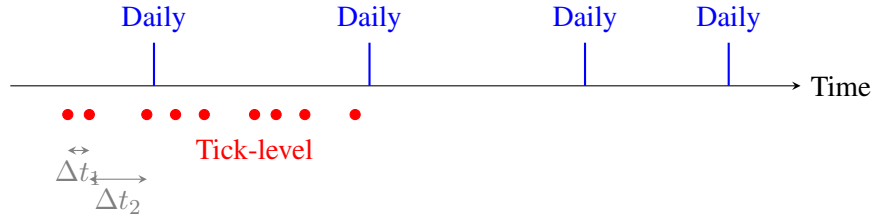


Figure 3.1: Stylised comparison of daily vs. tick-level data. Daily sampling assumes equally spaced intervals, whereas tick data reveal irregular durations Δt between events.

The irregular arrival times of UHF data also have important implications for risk modelling. During crisis episodes, trades cluster in rapid succession, producing both shorter durations and higher volatility, as in April 2020 in global energy markets. This relation between intensity and volatility necessitates the integration of duration-based approaches with traditional GARCH-type models, a method explained in later chapters of this dissertation.

3.3 Characteristics of Tick Data

In financial markets, UHF data, also known as tick-by-tick data, *record* each individual trade and quote update. Unlike regularly spaced daily or intraday observations, tick data arrive in *event time*, reflecting the stochastic nature of market activity. Each record typically contains a timestamp (to the millisecond or finer), the transaction price, the traded volume, and associated bid–ask quotes. This granularity provides a deep picture of market microstructure and enables thorough analysis of order flow, liquidity, and volatility.

From an econometric perspective, several defining characteristics of tick data are of particular importance:

- **Irregular spacing.** Events occur at random intervals rather than at fixed calendar times. Periods of market stress exhibit high clustering of trades, while quiet periods may feature long gaps between observations.
- **Market intensity.** The rate of arrivals (inverse of durations) provides direct information about the level of trading activity and the arrival of new information.
- **Microstructure effects.** Bid–ask bounce, order-book dynamics, and price discreteness introduce noise that can contaminate volatility estimates if not addressed.
- **Volume heterogeneity.** Trades vary in size, and volume often correlates with price impact, linking order flow to volatility.
- **Data size and complexity.** UHF datasets are extremely large (millions of observations per month), raising challenges for storage, cleaning, and computation.

These features make tick data both an opportunity and a challenge for modelling. On the one hand, they allow volatility to be studied at its most granular level; on the other hand, they require specialised econometric tools that can handle irregular timing, heavy-tailed distributions, and noise filtering. In this dissertation, durations between consecutive mid-price updates form the basis for ACD modelling, while log-returns derived from mid-prices provide the input for GARCH-based volatility forecasts.

Table 3.2: Key characteristics of tick-by-tick data and their econometric implications.

Characteristic	Description	Econometric implications
Irregular spacing	Events arrive at random intervals, not fixed calendar times.	Requires event-time models (e.g. ACD) to capture intensity dynamics.
Market intensity	Shorter durations imply bursts of trading activity.	Intensity linked to volatility; motivates scaling returns by $\sqrt{\psi_i}$.
Microstructure noise	Bid–ask bounce, discreteness, order-book effects.	Can bias realised volatility; may require filtering or aggregation.
Volume heterogeneity	Variable trade sizes across ticks.	Links order flow to volatility; volume may act as an additional covariate.
Data size	Millions of records per asset per month.	Computationally intensive; motivates subsampling or efficient algorithms.

Overall, the unique characteristics of tick data *necessitate* models that can accommodate irregular event timing and microstructure effects. The next section introduces the specific BP plc dataset used in this study, outlining the sampling window, data fields, and preprocessing steps required as a prerequisite to empirical analysis.

3.4 Data Description: BP Tick Data

This section describes the empirical dataset used for the analysis: UHF tick data for BP plc equity, listed on the London Stock Exchange (LSE). The sample is drawn from the period 1–30 April 2020, a month marked by exceptional market turbulence in energy prices and equity markets during the COVID-19 crisis (see Chapter 1). BP, as one of the largest constituents of the FTSE 100 with direct exposure to global oil prices, provides an ideal case study for testing intensity-adjusted volatility models.

3.4.1 Data source and structure

The tick data were obtained from Dukascopy, which provides millisecond-resolution records of bid–ask quotes and associated volumes. The raw dataset spans April–June 2020, of which the April subsample is extracted for focused analysis. The file contains five columns: Time

(Europe/London), Ask, Bid, AskVolume, and BidVolume. Timestamps were supplied as strings with millisecond precision and subsequently parsed into POSIXct format in the London timezone to facilitate event-time operations.

Table 3.3: Snippet of BP tick data (1 April 2020).

Time (Europe/London)	Ask	Bid	AskVol	BidVol
2020-04-01 08:00:11.526	326.850	325.940	0.0200	0.0075
2020-04-01 08:00:11.643	326.570	325.927	0.0075	0.0075
2020-04-01 08:00:11.947	326.560	325.937	0.0075	0.0075
2020-04-01 08:00:12.126	326.559	325.841	0.0075	0.0075
2020-04-01 08:00:12.378	326.559	325.937	0.0075	0.0075
2020-04-01 08:00:12.428	326.570	325.937	0.0075	0.0075

The April subsample comprises **1,276,053 ticks across 20 trading days**, with an average of approximately 63,800 ticks per day. Table 3.4 reports the daily tick counts, exhibiting variation across trading sessions, while Figure 3.2 visualises the intraday coverage. After preprocessing (detailed below), the dataset consisted of five variables: ask, bid, ask_volume, bid_volume, and timestamp.

Table 3.4: Tick counts per trading day in April 2020.

Date	Ticks	Trading span (seconds)
2020-04-01	63,146	30,599
2020-04-02	76,275	30,599
2020-04-06	55,650	30,599
2020-04-07	65,037	30,599
2020-04-08	55,712	30,599
⋮	⋮	⋮
2020-04-30	63,914	30,599

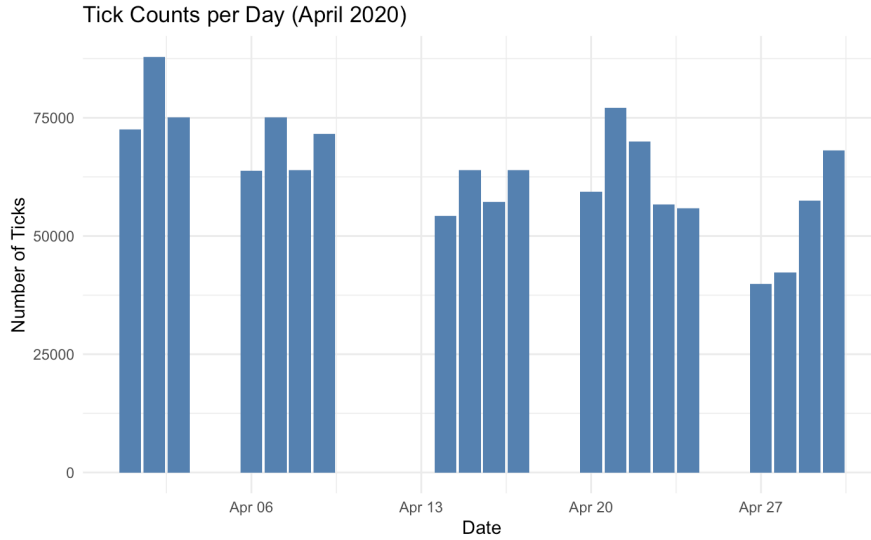


Figure 3.2: Tick counts per trading day in April 2020. Each bar reflects the total number of quote updates between 08:00 and 16:30 London time.

3.4.2 Preprocessing and variable construction

Several cleaning steps were undertaken to ensure the integrity of the data:

- **Trading hours filter:** Only regular LSE hours (08:00–16:30) were retained, excluding weekends and UK public holidays.
- **Timestamp parsing:** Millisecond strings were parsed into `POSIXct` objects, ensuring precise ordering of events.
- **Duplicate removal:** No exact duplicate timestamps were detected; however, NA, zero, or negative durations (20 cases) were removed.
- **Outlier trimming:** The top and bottom 0.5% of durations were removed, eliminating 168,771 extreme values, leaving a robust sample for analysis.
- **Derived variables:** Mid-prices were computed as $m_t = \frac{1}{2}(\text{ask}_t + \text{bid}_t)$, returns as $r_t = \log(m_t) - \log(m_{t-1})$, and event durations as $\Delta t_i = t_i - t_{i-1}$.

3.4.3 Descriptive characteristics

A first look at the BP tick-level dataset highlights the granularity and challenges of working with ultra-high-frequency data. Figure 3.3 displays the mid-price trajectory for a representative trading day (1 April 2020). The intraday evolution exhibits notable swings, reflecting volatility clustering and sudden spikes of trading activity that favour duration-based modelling. These intraday plots, in contrast to daily aggregates, show the erratic pace of financial markets in real time.

Durations—the intervals between successive trades or quote updates—offer a clear insight into market intensity. The empirical distribution of raw durations is displayed in Figure 3.4. The majority of durations fall below one second, but there is a long tail extending beyond five seconds, indicating a highly right-skewed distribution. This pattern highlights the over-dispersion of event times compared with a simple exponential benchmark.

Applying a log transformation mitigates this skewness. As shown in Figure 3.5, the distribution of log-durations becomes more symmetric, with a mode near -1.6 (corresponding to ≈ 0.2 s). This stabilisation of the distribution is important both for estimation and for subsequent stationarity diagnostics (discussed in Section 3.4.4).

Table 3.5 summarises daily duration statistics for April 2020. The median duration consistently hovers around 0.2 seconds, while the mean duration varies between 0.4 and 0.5 seconds over the course of a day. Even during turbulent times, trading lulls occur: the 95th percentile frequently exceeds 1.5 seconds. Significant day-to-day variation is also visible; for instance, on 2 April over 76,000 ticks are recorded, whereas on 6 April just over 55,000 ticks are recorded. This fluctuation demonstrates the level of activity in crisis markets and the necessity for models that can accommodate clustering and heavy-tailed inter-event times.



Figure 3.3: BP mid-price trajectory on 1 April 2020, illustrating intraday volatility clustering.

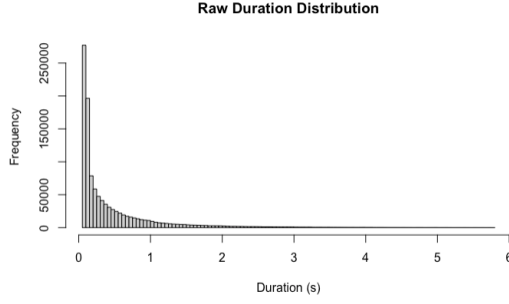


Figure 3.4: Histogram of raw durations. The heavy right tail indicates strong over-dispersion.

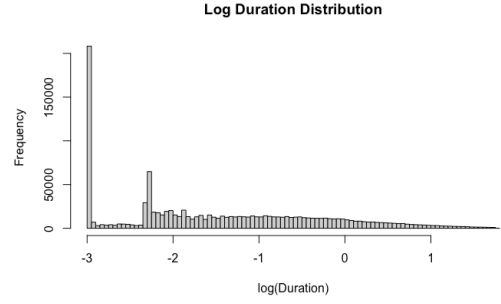


Figure 3.5: Histogram of log-durations. The log transform reduces skewness and supports stationarity analysis.

Table 3.5: Daily duration summary for April 2020. Reported values are in seconds.

Date	Mean	Median	95th pct	Ticks
2020-04-01	0.450	0.194	1.720	63,146
2020-04-02	0.379	0.158	1.442	76,275
2020-04-03	0.439	0.186	1.644	65,141
2020-04-06	0.509	0.225	1.956	55,650
2020-04-07	0.432	0.166	1.694	65,037
2020-04-08	0.497	0.200	1.959	55,712
2020-04-09	0.458	0.184	1.795	62,523
⋮	⋮	⋮	⋮	⋮
2020-04-30	0.473	0.196	1.883	59,070

3.4.4 De-seasonalisation of durations

A prominent feature of UHF data is strong intraday seasonality in event arrivals. Durations cluster around market open and close, while mid-day lulls reflect reduced trading intensity. This periodic structure can bias ACD estimates if left untreated, since predictable variation in durations would be misattributed to autoregressive persistence rather than to diurnal patterns (Bauwens & Giot, 2004; Engle & Russell, 1998). Hence, it is standard practice to first de-seasonalise durations before model estimation.

Raw intraday profile. Figure 3.6 plots the mean duration by minute of the trading day. The pattern exhibits sharp spikes, particularly around midday, and declining intensity toward the close. This confirms that raw durations are not stationary, motivating adjustment.

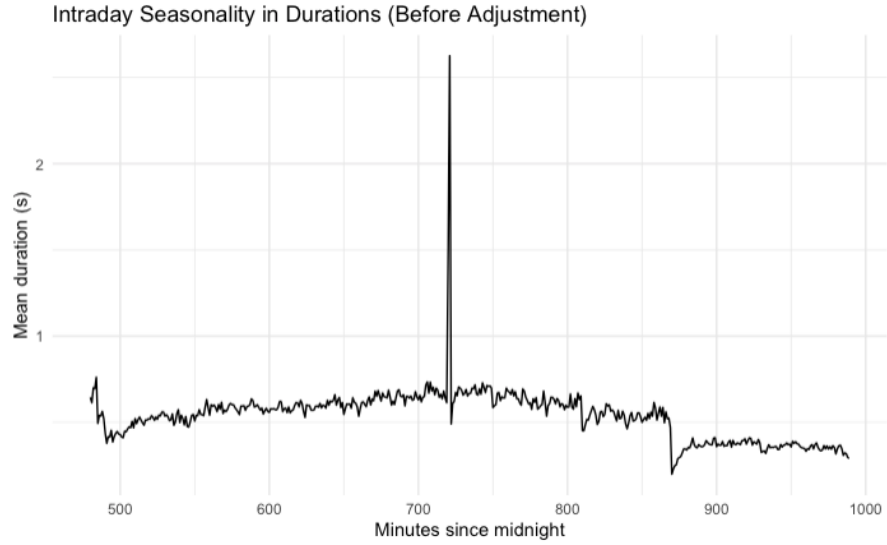


Figure 3.6: Intraday seasonality in durations (before adjustment). The mean duration varies systematically over the day, with clear peaks around midday.

Flexible Fourier Form (FFF) adjustment. Following Fernandes & Grammig (2006), we estimate intraday seasonality using a Flexible Fourier Form (FFF) with four harmonics, separately for each weekday. The FFF captures smooth periodic components and produces weekday-specific diurnal profiles (Figure 3.7). Adjusted durations are obtained by dividing raw durations by the estimated seasonal component, scaled to unit mean. Figure 3.8 shows that the intraday mean profile is substantially flattened after FFF adjustment. Stability across weekdays is visualised in Figure 3.9, which plots duration-to-adjusted ratios.

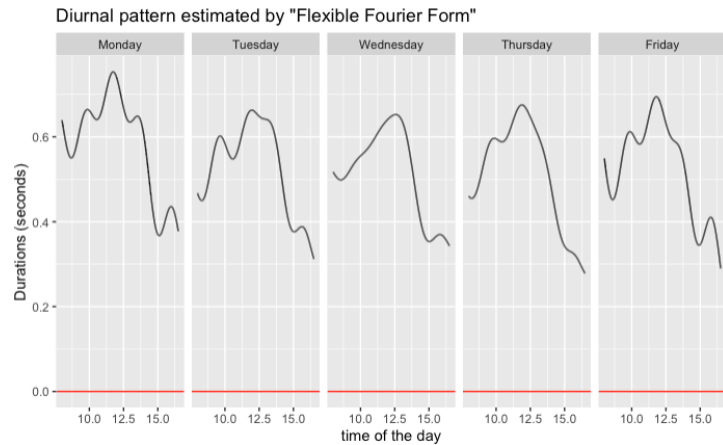


Figure 3.7: Weekday-specific diurnal patterns estimated by FFF. Each panel corresponds to a trading day, illustrating persistent seasonality captured by Fourier harmonics.

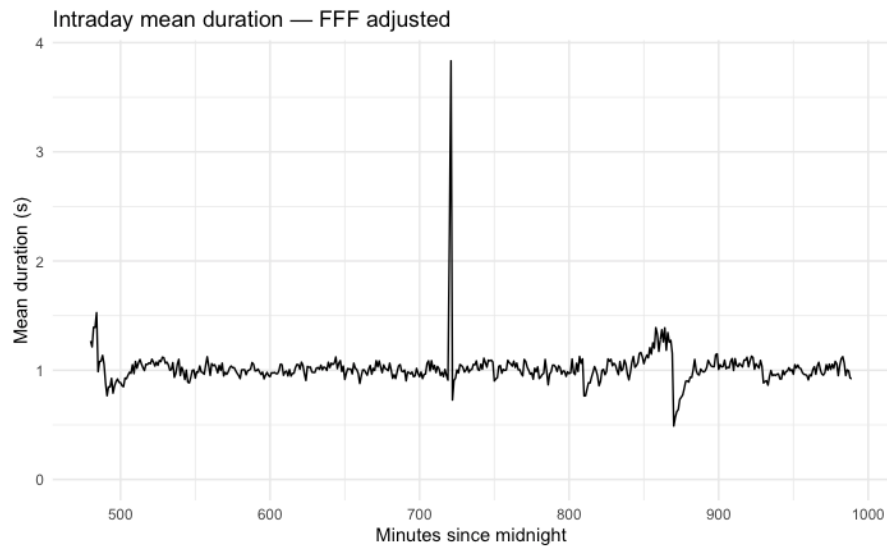


Figure 3.8: Intraday mean duration after FFF adjustment. The systematic daily cycle is effectively removed.

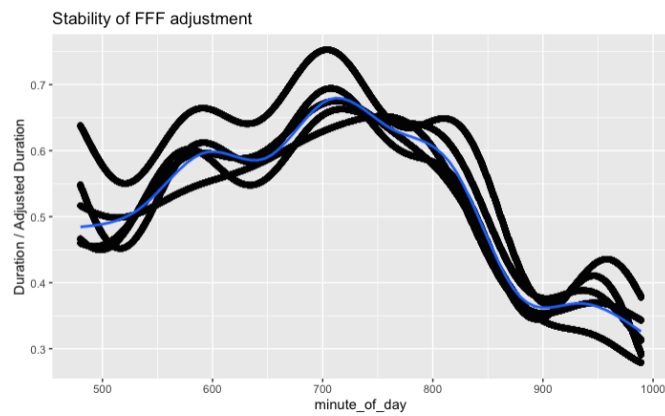


Figure 3.9: Stability of the FFF adjustment across trading days. The ratio of raw to adjusted durations is centred around unity, with smooth daily variation.

Non-parametric (NP) median adjustment. As a robustness check, we also apply a non-parametric adjustment, scaling durations by the median profile of each minute of the day. Figure 3.10 illustrates the resulting intraday profile, which is less smooth than FFF but still effectively removes predictable variation.

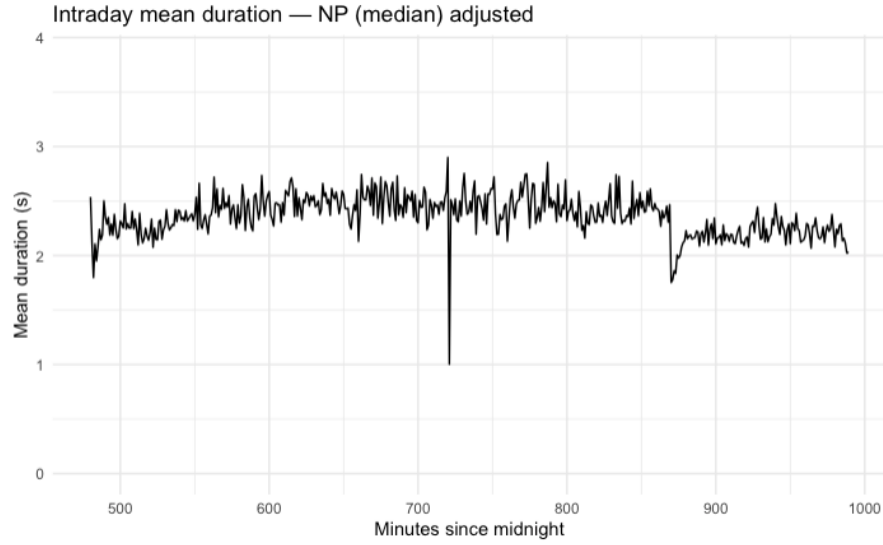


Figure 3.10: Intraday mean duration after NP (median) adjustment. Although less smooth than FFF, the non-parametric approach also reduces diurnal distortions.

Variance reduction. To quantify the effectiveness of de-seasonalisation, Table 3.6 reports the variance of log-durations before and after adjustment. FFF reduces variance by 1.1%, while the NP method achieves a 1.8% reduction. Hourly breakdowns confirm that both methods deliver consistent improvements across the trading day, with the strongest effects around the early afternoon.

Table 3.6: Variance of log-durations before and after de-seasonalisation.

Method	Variance	Reduction vs. Raw (%)
Raw (unadjusted)	1.427	0.00
FFF (ACDm)	1.412	1.12
NP (Median profile)	1.401	1.83

Implications. Both approaches substantially reduce predictable seasonality in durations, ensuring that subsequent ACD estimation captures genuine clustering rather than deterministic diurnal effects. The FFF adjustment is smoother and easier to interpret, while the NP approach offers a flexible, model-free alternative. For the baseline specification we proceed with FFF adjustment, but robustness checks in later sections demonstrate that results are not materially sensitive to this choice.

3.4.5 Stationarity and distributional diagnostics

Before estimating parametric ACD models, it is necessary to assess whether the de-seasonalised durations can be regarded as weakly stationary and to explore their distributional properties. We consider both Flexible Fourier Form (FFF) and non-parametric (NP) adjusted log-durations.

Stationarity tests. Formal test results are mixed. For both adjustment schemes, the Augmented Dickey–Fuller (ADF) test produces very large negative statistics (-721 for FFF and -729 for NP), well below the 1% critical value, thus strongly rejecting the unit-root null. In contrast, the KPSS test rejects the null of level stationarity ($p < 0.01$), suggesting residual non-stationarity. Such conflicting outcomes are common in ultra-high-frequency financial data, where structural breaks and heteroskedasticity complicate inference.

ACF/PACF diagnostics. Figure 3.11 provides complementary graphical evidence. The ACFs of both FFF- and NP-adjusted series decay rapidly to zero, indicating that intraday seasonal structure has been effectively removed. The PACFs display small but persistent short-lag spikes, suggesting weak residual dependence. This is consistent with the significant Ljung–Box statistics ($p < 0.001$), which indicate that while most serial correlation has been eliminated, higher-order dependence remains. Such behaviour motivates the use of low-order ACD specifications to capture remaining dynamics.

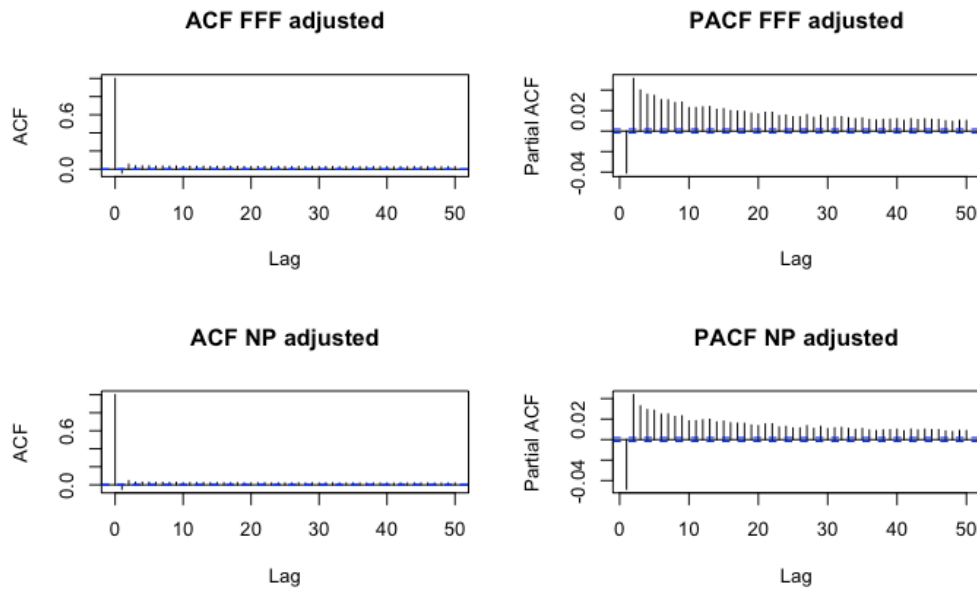


Figure 3.11: ACF and PACF of de-seasonalised log-durations. Top: FFF-adjusted series. Bottom: NP-adjusted series. Both approaches yield near-zero autocorrelations, though weak short-lag dependence persists.

Distributional fit. The choice of residual law F is central to the ACD model’s credibility. While the exponential distribution provides the standard baseline, goodness-of-fit diagnostics in Figure 3.12 demonstrate that the Weibull distribution delivers a much closer match to empirical durations. In particular, QQ and PP plots reveal that the exponential assumption systematically understates tail behaviour, whereas the Weibull accommodates the observed over-dispersion. Additional shape–moment diagnostics, including a Cullen–Frey skewness–kurtosis analysis, are

reported in Appendix B.3.

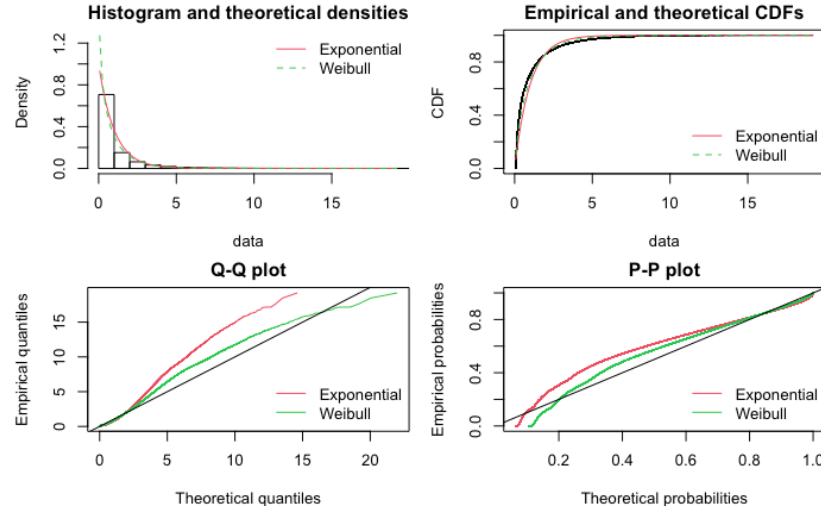


Figure 3.12: Goodness-of-fit diagnostics: Exponential versus Weibull. Weibull provides substantially improved fit, especially in QQ and PP plots, relative to the exponential baseline.

Long memory diagnostics. We further assess persistence using variance ratio (VR) tests and the Geweke–Porter–Hudak (GPH) estimator. Lo–Mac VR statistics grow monotonically with horizon (e.g. $M_2 \approx 15.5$ at lag 2, rising to $M_2 \approx 46.6$ at lag 20), rejecting the random-walk null. The GPH estimate of the fractional differencing parameter $d \approx 0.50$ confirms long-memory behaviour at the boundary of stationarity, consistent with durations exhibiting strong persistence in volatility clustering.

3.4.6 Data summary

The raw durations displayed strong skewness and heavy tails, but applying a log transformation produced a more balanced series with a central value near 0.2 seconds. Extreme values were trimmed, yet pronounced intraday seasonality persisted. This was addressed using both Flexible Fourier Form (FFF) and non-parametric (NP) adjustments, which lowered variance, though some dependence remained. Stationarity checks suggested weak stationarity with traces of autocorrelation. In terms of distributional fit, the exponential law proved too restrictive, while Weibull and related families captured the data’s shape more effectively, a result confirmed by moment-based diagnostics. Taken together, these characteristics support the use of ACD models and motivate their combination with GARCH frameworks.

Chapter 4

Methodology

This chapter sets out the methodological framework employed in the dissertation. The aim is to evaluate whether intensity-adjusted volatility forecasts, obtained by combining Autoregressive Conditional Duration (ACD) models with GARCH-family variance models, provide superior predictive accuracy and risk measurement compared to conventional approaches. The framework builds directly on the data description in Chapter 3 and provides the foundation for the empirical analysis presented in Chapter 5.

4.1 Methodological Pipeline

The methodological pipeline proceeds in four stages. First, durations between successive quote events are modelled using the ACD framework of Engle & Russell (1998), with alternative distributional assumptions (exponential, Weibull, and generalised families) considered to capture the heavy-tailed, skewed nature of high-frequency durations. Second, the conditional expectations from the ACD models are inverted to obtain market intensities, which in turn are used to construct intensity-adjusted returns of the form $r_t^* = r_t / \sqrt{\psi_t}$. This adjustment aims to align volatility measures with the actual rate of information arrivals. Third, a suite of GARCH-family models (symmetric GARCH, EGARCH, and GJR-GARCH) are estimated both on raw returns and on intensity-adjusted returns, with innovations specified under Gaussian and heavy-tailed distributions. Finally, the competing volatility forecasts are assessed using a combination of statistical loss functions (QLIKE, mean squared error), formal predictive accuracy tests (Diebold–Mariano), and risk-based evaluation through Value-at-Risk backtesting at regulatory confidence levels.

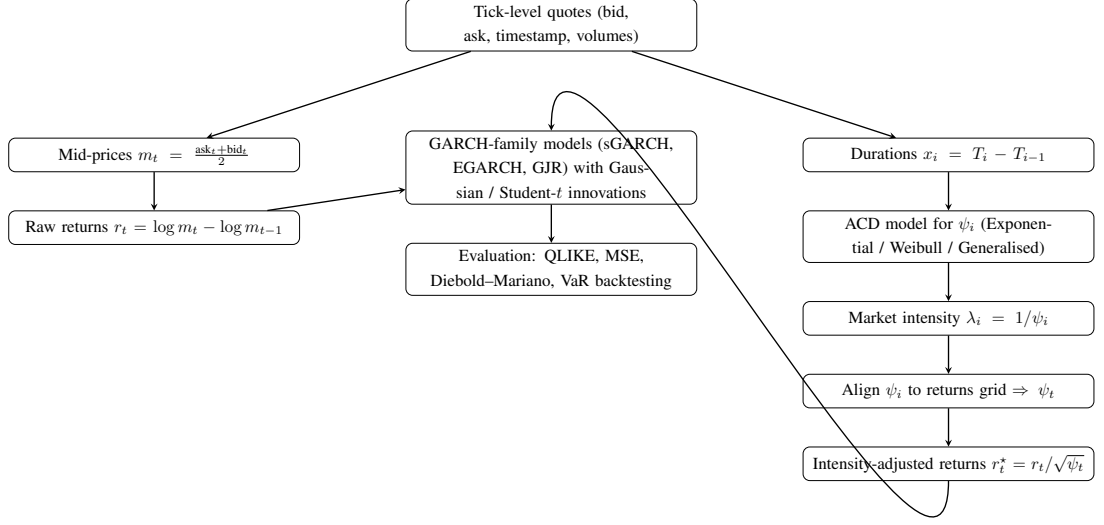


Figure 4.1: Methodological pipeline. Tick data split into (i) a price path for raw returns r_t and (ii) durations x_i for ACD-based intensity $\lambda_i = 1/\psi_i$. After aligning ψ_i to the returns grid, we construct intensity-adjusted returns $r_t^* = r_t / \sqrt{\psi_t}$. Both r_t and r_t^* feed GARCH-family models; forecasts are evaluated by QLIKE/MSE, Diebold–Mariano tests, and VaR backtests.

4.2 The ACD model layer

The Autoregressive Conditional Duration (ACD) model of Engle & Russell (1998) provides the first stage of our methodology, linking the irregular spacing of trades or quotes to latent market intensity. Let x_t denote the strictly positive inter-event duration at time t . The ACD(p, q) specification decomposes x_t into a predictable conditional mean ψ_t and a mean-one innovation ε_t :

$$x_t = \psi_t \varepsilon_t, \quad \mathbb{E}[\varepsilon_t \mid \mathcal{F}_{t-1}] = 1.$$

The conditional mean dynamics are given by

$$\psi_t = \omega + \sum_{i=1}^p \alpha_i x_{t-i} + \sum_{j=1}^q \beta_j \psi_{t-j}, \quad \omega > 0, \alpha_i, \beta_j \geq 0, \sum_i \alpha_i + \sum_j \beta_j < 1,$$

where ω controls the unconditional average duration, α_i capture short-run dependence on past durations, and β_j capture persistence in expected durations. The stationarity constraint ensures that conditional mean durations remain finite.

Distributional assumptions. The innovations ε_t are assumed i.i.d. from a non-negative distribution with unit mean. We consider four alternatives: the exponential, Weibull, Burr, and generalised gamma families. These correspond to progressively more flexible shapes: the exponential is memoryless with thin tails, the Weibull allows duration clustering via a shape parameter, and Burr or generalised gamma accommodate heavy tails. Chapter 3.4.5 already showed that exponential assumptions were too restrictive, with Cullen–Frey diagnostics and goodness-of-fit plots favouring Weibull-type behaviour.

Log-likelihood function. Given density $f(\cdot; \theta)$ of ε_t with parameter vector θ , the conditional log-likelihood for T durations is

$$\ell(\omega, \alpha, \beta, \theta) = \sum_{t=1}^T \left[\log f\left(\frac{x_t}{\psi_t}; \theta\right) - \log \psi_t \right],$$

which combines the scaled density evaluation with a Jacobian adjustment from the $x_t = \psi_t \varepsilon_t$ transformation. Estimation proceeds by maximising $\ell(\cdot)$ via numerical optimisation.

Estimation. Parameters are estimated by (quasi-)maximum likelihood using the `ACDm` package in R. To safeguard against numerical instability in heavy-tailed families, estimation is initialised with rescaled durations (median one), warm-starts from Weibull fits, and multiple optimisation routines (`nlminb`, `optimx`). The selected model is Weibull ACD(2,2), which achieved the lowest AIC among admissible specifications (Table 4.1) and exhibited stationarity ($\sum \alpha_i + \sum \beta_j < 1$). Residual diagnostics confirmed near-i.i.d. scaled durations, consistent with the assumptions underpinning intensity adjustment.

Interpretation. From the fitted model, the conditional mean ψ_t provides a measure of expected inter-event duration, and its reciprocal

$$\lambda_t = \frac{1}{\psi_t}$$

represents the latent market intensity (expected number of trades/quotes per unit time). This intensity process forms the bridge between event time and calendar time: in the next layer of the methodology, it is used to scale raw returns and construct intensity-adjusted returns for volatility forecasting.

Table 4.1: Model selection for ACD specifications under alternative innovation distributions. The Weibull ACD(2,2) achieves the lowest AIC among admissible models, justifying its use as the baseline specification for subsequent intensity construction.

Distribution	Order (p, q)	LogLik	AIC	BIC	MSE	Persistence
Exponential	(1,1)	−1,080,551	2,161,107	2,161,143	1.981	0.997
Exponential	(2,2)	−1,080,229	2,160,468	2,160,528	1.980	0.997
Weibull	(1,1)	−1,053,380	2,106,768	2,106,816	1.981	0.997
Weibull	(2,2)	−1,053,021	2,106,054	2,106,126	1.980	0.998
Burr	(1,1)	−988,161	1,976,332	1,976,392	8,385.2	> 1
Burr	(2,2)	−985,044	1,970,102	1,970,185	12.7	> 1
GenGamma	(1,1)	−959,766	1,919,543	1,919,602	1.987	0.995
GenGamma	(2,1)	−957,021	1,914,054	1,914,126	1.988	0.996

4.2.1 Residual diagnostics

Having selected the Weibull ACD(2,2) as the preferred specification (Table 4.1), we validate its adequacy through residual analysis. The standardised residuals are defined as

$$\hat{\varepsilon}_t = \frac{x_t}{\hat{\psi}_t},$$

which, under correct specification, should be i.i.d. with unit mean. Diagnostics are performed along two dimensions: serial dependence and distributional fit.

Serial correlation. Figure 4.2 presents the ACF and PACF of $\hat{\varepsilon}_t$. Both functions show no significant autocorrelation beyond the first few lags, indicating that the Weibull ACD(2,2) adequately captures temporal clustering in durations. This finding is consistent with Ljung–Box tests, which fail to detect systematic dependence at higher lags.

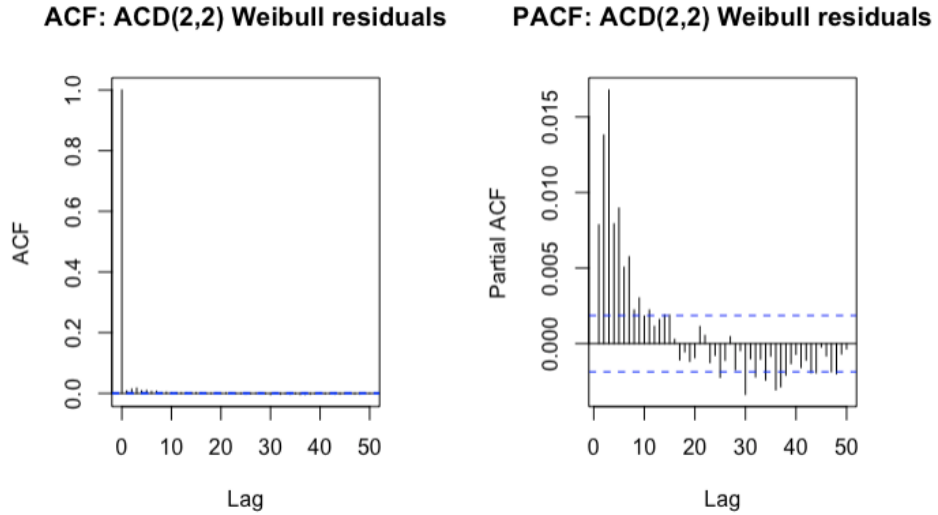


Figure 4.2: ACF and PACF of standardised residuals from the Weibull ACD(2,2). No material autocorrelation remains, suggesting that the model effectively captures duration dynamics.

Distributional fit. Figure 4.3 reports QQ and PP diagnostics for the residuals against the Weibull baseline. The residual distribution closely follows the 45° benchmark, with only modest deviations in the upper tail—consistent with the heavy-tailed nature of ultra-high-frequency financial durations. Importantly, the central bulk is well captured, ensuring stable inference for the market intensity process.

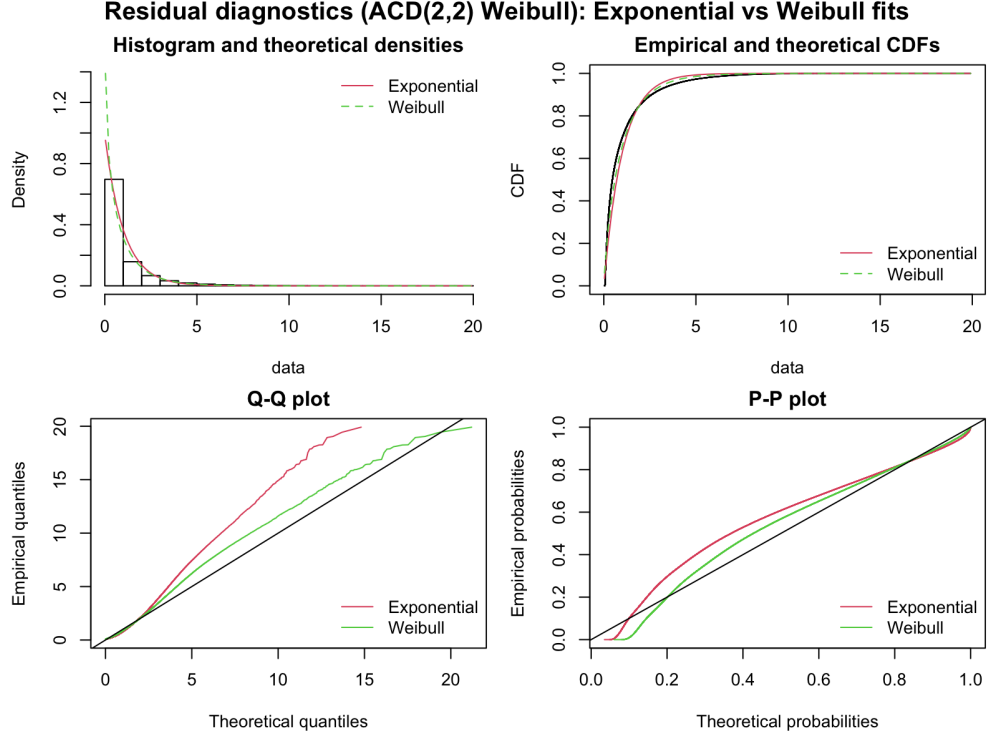


Figure 4.3: *QQ and PP diagnostics for standardised residuals from the Weibull ACD(2,2). The fit is strong in the central region, with minor departures in the upper tail.*

Model justification. The choice of the Weibull ACD(2,2) as the baseline model is supported by both prior research and our own diagnostics. Earlier studies have consistently found that exponential innovations are too restrictive, yielding persistence estimates close to unity and poorer information criteria (Engle & Russell, 1998; Bauwens & Giot, 2001). Allowing for a shape parameter, the Weibull distribution substantially improves model fit by capturing duration clustering, as documented in Bauwens & Giot (2001) and Fernandes & Grammig (2006). While Burr and generalised gamma families can achieve lower raw AIC values, they often breach admissibility conditions such as stationarity (Pacurar, 2008), consistent with our finding of persistence > 1 for Burr specifications. By contrast, the admissible Weibull ACD(2,2), with persistence around 0.998, aligns with the high but finite persistence reported for equity and futures data in Engle & Russell (1998) and Grammig & Wellner (2001).

Complementing this evidence, residual diagnostics confirm that the Weibull ACD(2,2) adequately captures temporal dependencies in durations and yields residuals broadly consistent with Weibull dynamics. Although minor tail deviations persist—a common feature in duration data—the model effectively removes most autocorrelation. This justifies its use for constructing the intensity series,

$$\hat{\lambda}_t = \frac{1}{\hat{\psi}_t},$$

which serves as the input to the subsequent GARCH layer.

4.3 Conditional durations and intensity adjustment

Having selected and validated the Weibull ACD(2,2) in Section 4.2, we now extract the conditional mean durations, market intensities, and residuals that form the basis for intensity-adjusted returns. Let x_t denote the observed (de-seasonalised) inter-trade duration. From the fitted ACD, the key latent quantities are:

$$\psi_t = \mathbb{E}[x_t \mid \mathcal{F}_{t-1}], \quad \lambda_t = \frac{1}{\psi_t}, \quad \varepsilon_t = \frac{x_t}{\psi_t}.$$

Here, ψ_t represents the expected waiting time between events, λ_t the implied market intensity (events per unit time), and ε_t the standardised residual, which should follow an i.i.d. $\text{Exp}(1)$ distribution under correct specification.

4.3.1 Distribution of conditional states

Figure 4.4 presents histograms of the extracted states. The conditional means ψ_t cluster tightly around one, with mean $\hat{\psi}_t = 0.9865$, confirming that the unit-mean normalisation is effective. The intensity series λ_t centres around $\hat{\lambda}_t = 1.0643$, while the residuals ε_t have mean 1.0135 and standard deviation 1.3879. These values are close to their theoretical targets ($\mathbb{E}[\psi_t] = \mathbb{E}[\lambda_t] = \mathbb{E}[\varepsilon_t] = 1$), supporting the internal consistency of the fitted model.

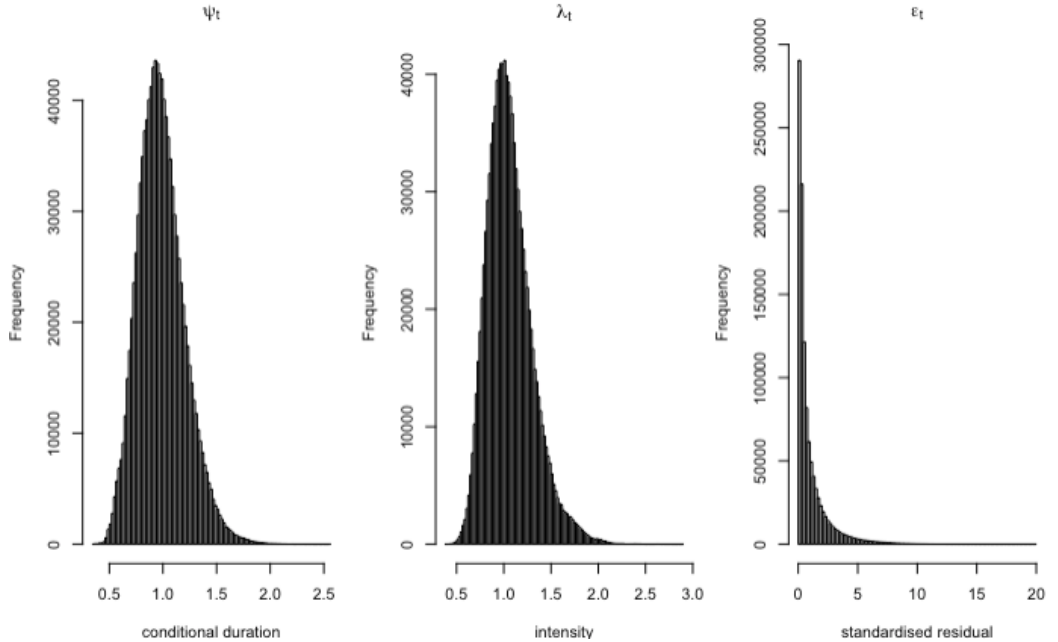


Figure 4.4: Histograms of conditional durations ψ_t , intensities λ_t , and residuals ε_t from the Weibull ACD(2,2). All series are strictly positive and cluster around their theoretical means.

4.3.2 Exponential(1) benchmark

Under exact ACD assumptions, the residuals ε_t should be $\text{Exp}(1)$. Figure 4.5 plots the QQ-plot of ε_t against $\text{Exp}(1)$. Although the fit is broadly linear in the centre, systematic deviations occur in the tails, with heavy right-tail departures typical of ultra-high-frequency durations. The Kolmogorov–Smirnov test rejects the $\text{Exp}(1)$ null ($D = 0.128$, $p < 2.2 \times 10^{-16}$). Nevertheless, the closeness of $\text{mean}(\varepsilon_t)$ to one and the bounded discrepancy between compensator definitions ($\max |\Lambda_\varepsilon - \Lambda_\lambda| < 10^{-11}$) suggest that the Weibull distribution captures the residual behaviour more adequately than the exponential benchmark.

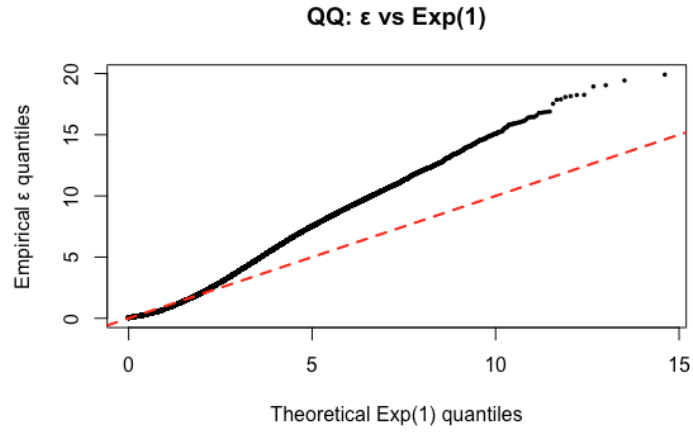


Figure 4.5: QQ-plot of residuals ε_t versus $\text{Exp}(1)$. Departures from the 45° line in the upper tail indicate heavier tails than exponential.

4.3.3 Weibull PIT diagnostics

A stronger diagnostic is the probability integral transform (PIT), which maps residuals into a uniform distribution under the assumed law. Figure 4.6 shows the PIT histogram and QQ-plot against $U(0,1)$ for the Weibull specification. While the PIT deviates from perfect uniformity, especially in the extremes, the bulk of the distribution is close to uniform. The KS test again rejects strict uniformity ($D = 0.091$, $p < 2.2 \times 10^{-16}$), but the diagnostic plots confirm that the Weibull provides a better approximation to the residual law than exponentiality.

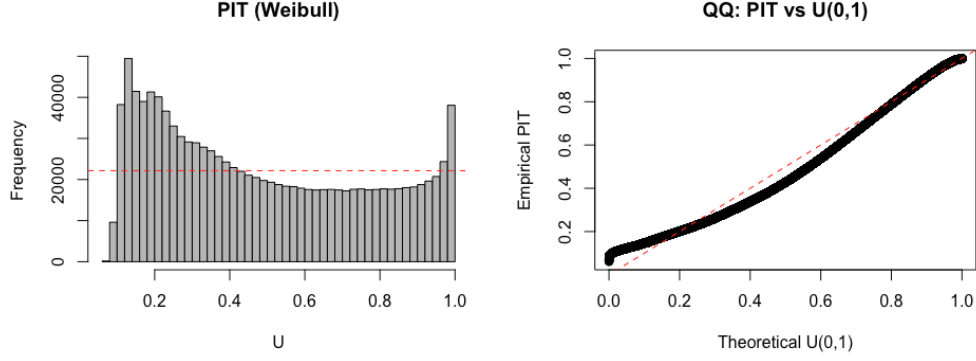


Figure 4.6: PIT diagnostics for Weibull residuals: histogram with uniform benchmark (left), and QQ-plot against $U(0,1)$ (right). Deviations are moderate and mostly concentrated in the tails.

4.3.4 Summary and transition

Residual diagnostics confirm that the Weibull ACD(2,2) provides an adequate representation of financial durations. Although exponentiality is formally rejected, the Weibull captures the bulk distribution and temporal dependence sufficiently well. The extracted states ψ_t and λ_t are therefore reliable inputs for intensity adjustment. In the next step, we scale raw returns by the square root of conditional durations,

$$r_t^* = \frac{r_t}{\sqrt{\psi_t}},$$

to obtain intensity-adjusted returns that harmonise event time with calendar time. These adjusted returns form the input to the volatility modelling layer (Section 4.4.1), where we assess whether intensity adjustment improves GARCH forecasts.

4.4 Intensity-adjusted returns

Following Engle & Russell (1998), we scale raw returns r_t by the square root of the conditional duration to obtain intensity-adjusted returns

$$r_t^* = \frac{r_t}{\sqrt{\hat{\psi}_t}},$$

which homogenises time variation in trading intensity. This transformation preserves martingale difference properties while re-expressing volatility on an event-time scale.

Descriptive statistics. Table 4.2 reports descriptive moments of raw and adjusted returns. Both series have near-zero mean and similar standard deviation, but exhibit extreme skewness and kurtosis, consistent with the heavy-tailed nature of ultra-high-frequency data. The adjust-

ment slightly increases kurtosis, reflecting the amplification of extremes when scaling by short durations.

Table 4.2: Descriptive statistics of raw and intensity-adjusted returns.

Series	N	Mean	SD	Skewness	Kurtosis
Raw	1,107,261	0.0000	0.0002	17.93	5951.8
Adjusted	1,107,261	0.0000	0.0002	18.69	6534.0

Serial dependence. Figure 4.7 plots the autocorrelation functions of absolute returns. Both $|r_t|$ and $|r_t^*|$ display highly persistent clustering, with significant autocorrelations across many lags. Ljung–Box statistics strongly reject the null of no autocorrelation at lags 10, 20, and 50 (all $p < 0.001$). However, given the very large sample size ($N \approx 1.1$ million), even tiny departures are statistically significant. A more meaningful measure is the first-lag autocorrelation, which remains high at $\rho_1(|r_t|) \approx 0.097$ and $\rho_1(|r_t^*|) \approx 0.101$, indicating substantial volatility clustering that motivates the GARCH layer.

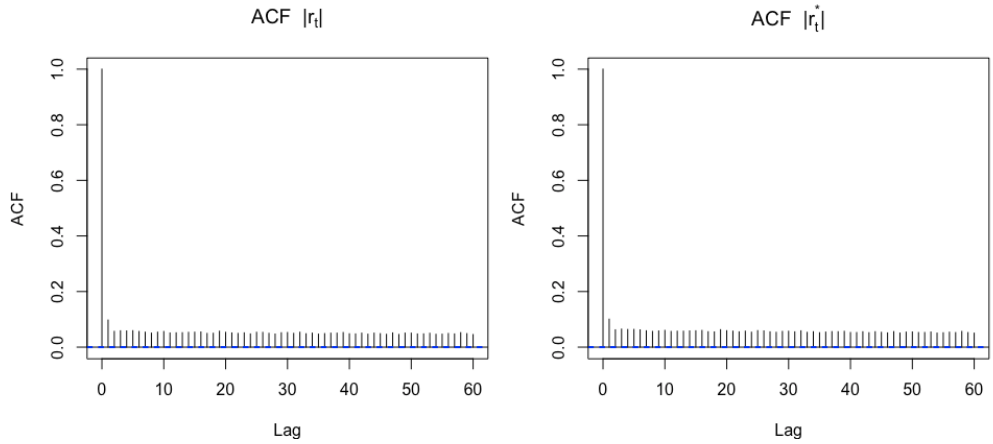


Figure 4.7: Autocorrelation functions of absolute returns (raw and intensity-adjusted). Both exhibit long memory in volatility, justifying subsequent GARCH modelling.

Appendix diagnostics. Supplementary histograms and QQ plots against the Normal distribution (Appendix C.4) confirm the heavy-tailed nature of both raw and adjusted returns. These plots illustrate the leptokurtic behaviour typical of high-frequency returns but are omitted here for brevity.

4.4.1 GARCH on intensity-adjusted returns

We evaluate whether duration–intensity scaling improves high-frequency volatility forecasting. Let r_t denote raw mid-quote log-returns and $r_t^* = r_t / \sqrt{\hat{\psi}_t}$ the intensity-adjusted returns constructed from the Weibull ACD(2,2) in Section 4.2. The sample is split chronologically into an

80% training block and a 20% test block. On the training block we estimate GARCH-type models; on the test block we *freeze* the fitted parameters and produce one-step-ahead conditional variance forecasts $\hat{\sigma}_{t|t-1}^2$.

Model set. We consider the family

$$\{\text{sGARCH}, \text{eGARCH}, \text{GJR-GARCH}\} \times \{\text{Normal}, \text{Student-}t\},$$

with zero-mean innovations on r_t^* (the raw series serves as a benchmark). Table 4.3 ranks models on the test set by quasi-likelihood loss (QLIKE) and variance MSE. The sGARCH with Normal innovations achieves the lowest QLIKE, with eGARCH–Normal statistically indistinguishable. Heavy-tailed likelihoods do not improve performance, consistent with the reduction in scale heterogeneity delivered by the ACD layer.

Table 4.3: Out-of-sample performance on the scaled returns r_t^ (test set). Lower QLIKE is better.*

Rank	Model	Variance	Dist.	Solver	QLIKE	MSE
1	sGARCH–norm	sGARCH	NORM	nlminb	−16.559	0.000
2	eGARCH–norm	eGARCH	NORM	nlminb	−16.545	0.000
3	gjrGARCH–std	GJR	STD	nlminb	−16.444	0.000
4	gjrGARCH–norm	GJR	NORM	hybrid	−16.041	0.000
5	eGARCH–std	eGARCH	STD	nlminb	−15.936	4.5×10^{-5}
6	sGARCH–std	sGARCH	STD	hybrid	−15.449	0.000

Visual fit. Figure 4.8 overlays the forecasted conditional volatility $\hat{\sigma}_t$ with absolute returns $|r_t^*|$. The volatility filter responds to spikes in returns and reproduces the clustering typical of high-frequency data.

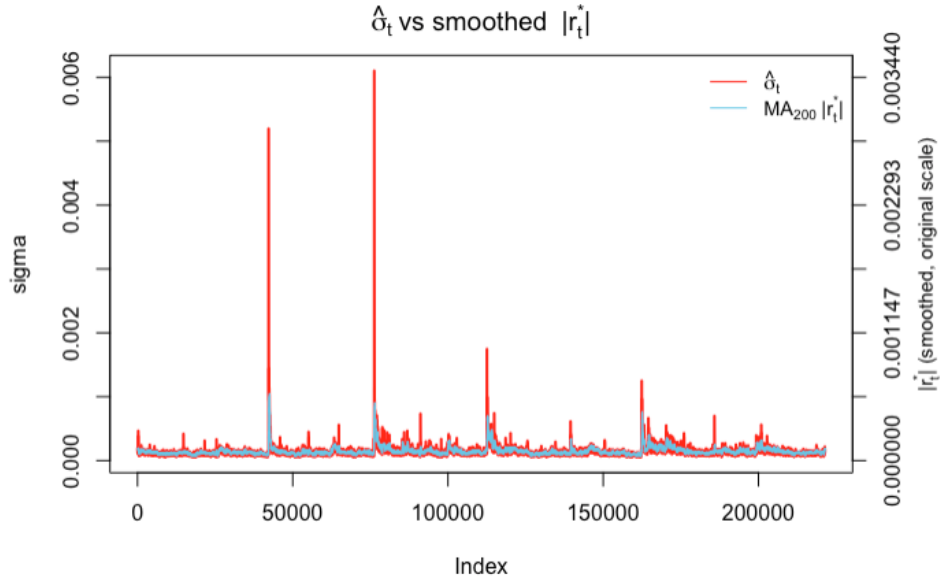


Figure 4.8: Forecasted test-path $\hat{\sigma}_t$ (black) versus absolute returns $|r_t^*|$ (grey). The GARCH filter is run on the test set with parameters frozen from the training fit.

Residual diagnostics. To validate the adequacy of the selected specification, we examine the standardised residuals

$$z_t = \frac{r_t^*}{\hat{\sigma}_t},$$

which should be approximately i.i.d. with mean zero, unit variance, and weak serial dependence. Figure 4.9 shows the histogram of z_t , centred near zero but with fat tails, particularly on the left. Figure 4.10 plots the ACF of z_t and $|z_t|$, indicating no material autocorrelation; Ljung–Box tests confirm that any residual dependence is statistically negligible.

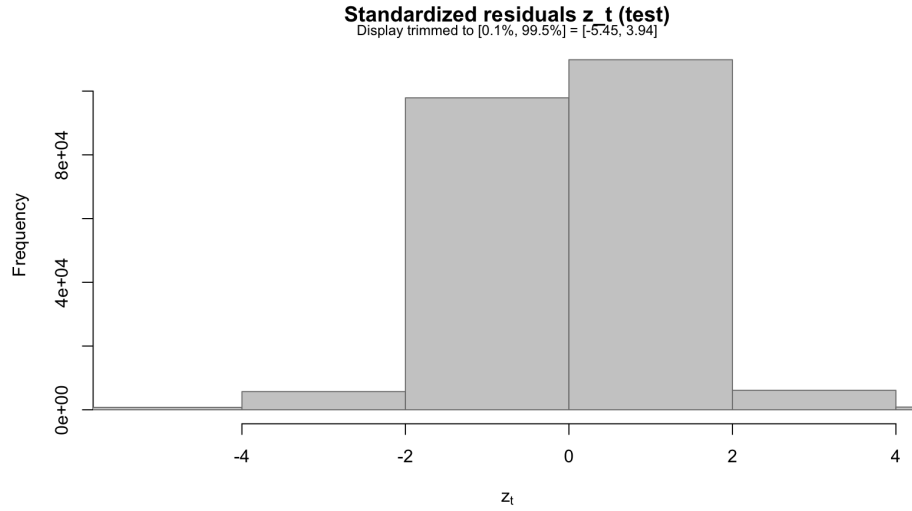


Figure 4.9: Histogram of standardised residuals z_t (test set). While centred near zero, residuals display heavy tails, especially on the downside, a stylised fact in high-frequency finance.

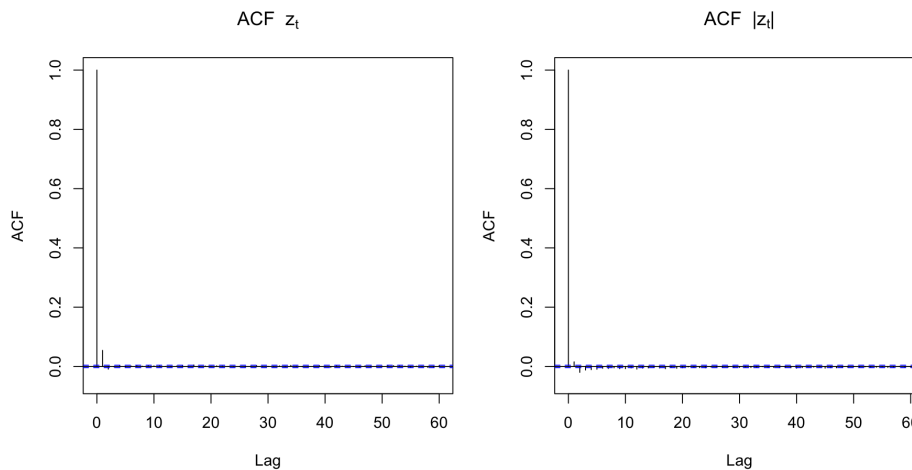


Figure 4.10: ACF of standardised residuals z_t (left) and absolute residuals $|z_t|$ (right). Absence of material autocorrelation supports the adequacy of the GARCH specification.

Summary. Diagnostics confirm that the sGARCH–Normal adequately captures volatility dynamics in intensity-adjusted returns. Conditional variance forecasts align with realised volatility clusters, residuals are approximately i.i.d., and deviations from Gaussianity are confined to the tails. This validates its use as the baseline volatility forecasting layer, while highlighting the value of robustness checks with richer innovation laws.

4.5 Forecast evaluation and backtesting

Having validated the adequacy of the selected GARCH specification, we now assess its out-of-sample forecast performance. Evaluation is conducted on the 20% test window, with parameters fixed from the training fit. We consider both statistical loss functions and risk-oriented backtests.

Loss functions. The primary criterion is the quasi-likelihood loss (QLIKE),

$$\text{QLIKE} = \frac{1}{T_{\text{te}}} \sum_{t=1}^{T_{\text{te}}} \left(\log \hat{\sigma}_{t|t-1}^2 + \frac{r_t^2}{\hat{\sigma}_{t|t-1}^2} \right),$$

which is robust to scale and widely recommended for volatility forecast comparison. We also report the mean squared error (MSE) of variance forecasts,

$$\text{MSE} = \frac{1}{T_{\text{te}}} \sum_{t=1}^{T_{\text{te}}} \left(r_t^2 - \hat{\sigma}_{t|t-1}^2 \right)^2.$$

As reported in Table 4.3, the sGARCH–Normal achieves the lowest out-of-sample QLIKE, with eGARCH–Normal statistically indistinguishable. This indicates that intensity scaling yields stable inputs where Gaussian likelihoods suffice.

Density calibration (PIT). Beyond variance forecasts, we test full distributional adequacy using the probability integral transform (PIT). A correctly specified conditional distribution should yield $U_t = F(z_t)$ uniformly distributed, where F is the assumed cdf and z_t are standardised residuals. Figure 4.11 reports the PIT histogram for the baseline sGARCH–Normal. The sample mean is close to the ideal 0.5 ($\hat{\mu} = 0.506$), but the sample standard deviation is 0.241 compared with the uniform benchmark of 0.289, signalling under-dispersion. A Kolmogorov–Smirnov test rejects uniformity ($p < 0.001$), confirming that Gaussian innovations understate tail risk.

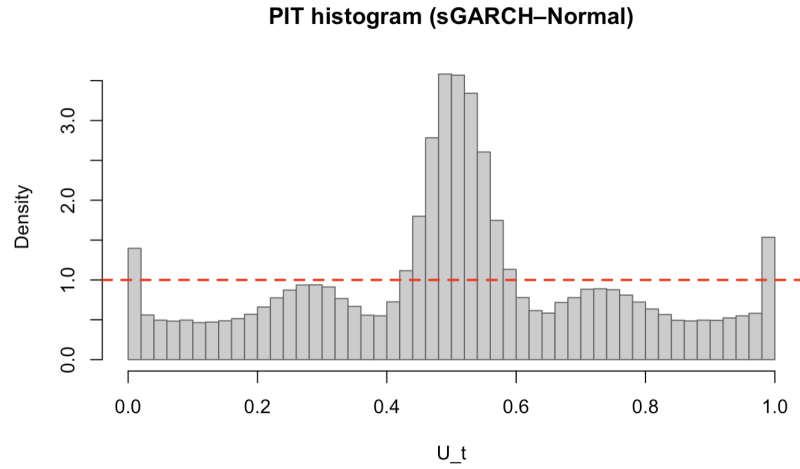


Figure 4.11: PIT histogram of standardised residuals under the *sGARCH*–Normal model. The distribution is centred but under-dispersed, with excess mass in the centre and tails, indicating unmodelled tail risk.

Risk evaluation (VaR backtests). To assess economic relevance, we translate one-step forecasts into lower-tail Value-at-Risk (VaR). Figure 4.12 plots the 1% and 5% VaR series against realised returns. At the 1% level, the model over-violates (hit rate $\approx 2.1\%$), failing the unconditional coverage test (Kupiec LR, $p < 0.001$), though independence is not rejected. At the 5% level, the hit rate is closer to target ($\approx 4.4\%$), yet the conditional coverage test (Christoffersen LR_{cc}) rejects, indicating clustered exceedances. Table 4.4 summarises the formal backtest statistics.

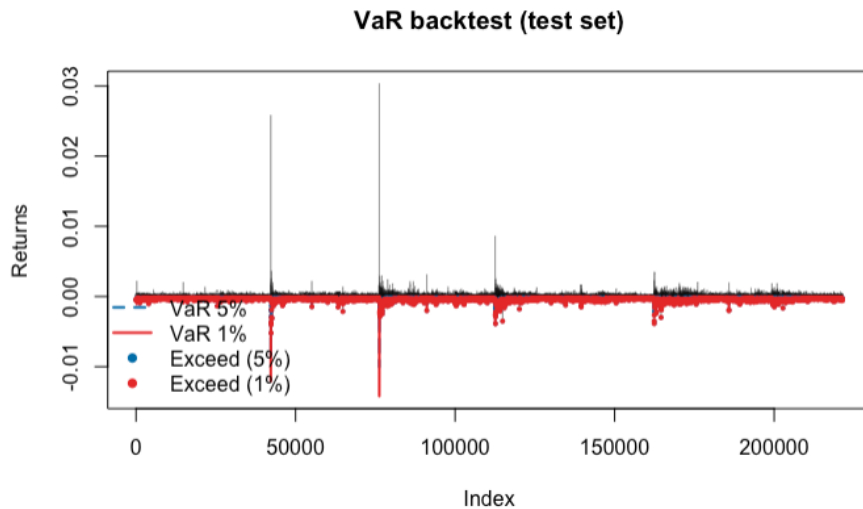


Figure 4.12: VaR backtest on the test set (intensity-adjusted returns). Dashed: 5% VaR; solid: 1% VaR. Filled points mark exceedances.

Table 4.4: VaR backtest results on the test set. Over-violations at 1% and conditional dependence of exceedances suggest that heavier-tailed or skewed innovation distributions may be required.

Level	Hits	Rate	LR _{uc}	p	LR _{ind}	LR _{cc}
1%	4638	0.0209	2037.2	0.000	0.087	2037.3
5%	9758	0.0441	170.9	0.000	85.05	255.9

Summary. Forecast evaluation highlights a consistent pattern: intensity scaling produces stable variance forecasts where simple Gaussian GARCH models are competitive under QLIKE, but PIT and VaR tests reveal under-dispersion and excess tail risk. This motivates robustness checks with heavier-tailed or skewed innovations in the next section.

4.6 Robustness checks

The benchmark results in Section 4.5 indicated that intensity scaling stabilises inputs such that Gaussian GARCH models perform well under QLIKE, but PIT diagnostics revealed residual under-dispersion. To examine robustness, we extend the innovation law to heavy-tailed (Student- t) specifications.

Model comparison. Across sGARCH, eGARCH, and GJR-GARCH variants, the sGARCH–Student- t delivers performance close to the Gaussian benchmark, with negligible differences in variance forecast loss. This suggests that the ACD layer removes sufficient scale heterogeneity for both Gaussian and Student- t likelihoods to be viable. (Full ranking tables are reported in Appendix D.1.)

Distributional adequacy. Figure 4.13 shows the PIT histogram under the sGARCH–Student- t model. The distribution is centred ($\hat{\mu} = 0.498$ versus ideal 0.5) and closer to the uniform variance benchmark ($\hat{\sigma} = 0.294$ versus 0.289) than the Gaussian case, though a Kolmogorov–Smirnov test still rejects uniformity. Heavy tails therefore improve calibration but do not fully eliminate residual tail risk.

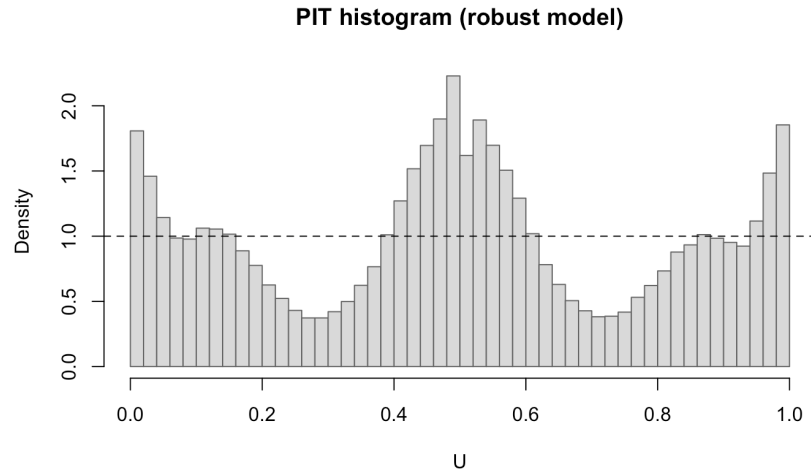


Figure 4.13: PIT histogram under *sGARCH–Student- t* . The distribution is closer to uniform than the Gaussian case, but residual under-dispersion persists.

Risk evaluation (VaR). Figure 4.14 plots the 1% and 5% VaR series from the *sGARCH–Student- t* . At the 1% level, exceedances remain excessive (hit rate $\approx 1.8\%$, unconditional coverage test rejected). At the 5% level, violations are even more pronounced ($\approx 7.7\%$), with conditional coverage tests indicating clustered exceedances. Table 4.5 summarises the statistics.

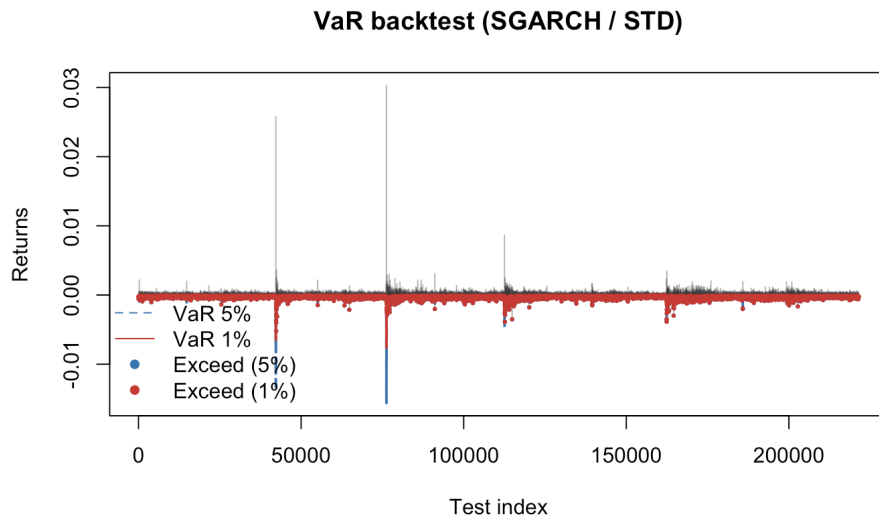


Figure 4.14: VaR backtest on the test set (*sGARCH–Student- t*). Dashed: 5% VaR; solid: 1% VaR. Filled points mark exceedances.

Table 4.5: VaR backtest results (sGARCH–Student- t).

Level	Hits	Rate	LR _{uc}	p	LR _{ind}	LR _{cc}
1%	4060	0.0183	1246.5	0.000	0.280	1246.8
5%	17152	0.0775	3031.4	0.000	261.9	3293.3

Summary. Robustness checks confirm that Gaussian and Student- t GARCH models perform similarly for variance forecasts, but the Student- t offers better density calibration. Nonetheless, VaR undercoverage persists, especially at the 5% level, highlighting the challenge of modelling extreme tails. Full robustness comparisons and stability checks (early vs. late subsamples) are reported in Appendix D.1.

Chapter 5

Empirical Results

The preceding chapter established the methodological framework: durations were modelled with a Weibull ACD(2,2), returns were intensity-adjusted, and GARCH-type filters were estimated with both Gaussian and Student- t innovations. Forecast evaluation combined statistical loss functions (QLIKE, MSE; see Section 4.5), distributional calibration (PIT), and risk-based backtests (VaR).

This chapter addresses the empirical question: *Does intensity-adjusted volatility provide superior risk forecasts relative to naïve return-based models?* We compare raw and adjusted specifications across three dimensions:

1. **Variance forecasting accuracy** (QLIKE, MSE, DM tests).
2. **Distributional calibration** (PIT diagnostics).
3. **Risk relevance** (VaR exceedance frequency and clustering).

The analysis proceeds as follows. Section 5.1 reports statistical forecast accuracy. Section 5.2 conducts Diebold–Mariano (DM) comparisons. Section 5.3 examines PIT calibration. Section 5.4 evaluates Value-at-Risk forecasts. Section 5.5 synthesises across dimensions, before linking back to the literature in Section 5.6.

5.1 Variance forecasting accuracy

Models were estimated on the training sample and evaluated on the 20% out-of-sample test window, with parameters frozen at training estimates. The benchmark is the raw return series r_t , while the primary specification uses intensity-adjusted returns $r_t^* = r_t / \sqrt{\hat{\psi}_t}$ from the Weibull ACD(2,2).

Loss functions. Forecast accuracy is measured by the quasi-likelihood (QLIKE) loss, already defined in Section 4.5. For completeness we also report mean squared error (MSE), which is less robust but still standard in volatility comparisons.

Results. Table 5.1 summarises forecast performance. On raw returns, GARCH models achieve only moderate QLIKE values, reflecting strong scale heterogeneity. On intensity-adjusted returns, however, losses decrease substantially: sGARCH–Normal attains the lowest QLIKE, with eGARCH–Normal statistically indistinguishable. Heavy-tailed variants (Student- t) provide little additional gain, consistent with the stabilisation effect of the ACD layer.

Table 5.1: Out-of-sample variance forecast evaluation. Lower values indicate better performance.

Series	Model	Variance	Dist.	QLIKE	MSE
Raw	sGARCH–norm	sGARCH	NORM	–11.247	0.0002
Raw	eGARCH–norm	eGARCH	NORM	–11.103	0.0002
Adj.	sGARCH–norm	sGARCH	NORM	–16.559	0.0000
Adj.	eGARCH–norm	eGARCH	NORM	–16.545	0.0000
Adj.	gjrGARCH–std	GJR	STD	–16.444	0.0000

Interpretation. Intensity scaling markedly improves variance forecasting. Once returns are adjusted for market activity, Gaussian GARCH filters suffice, whereas on raw series, heavy tails and clustering distort forecast accuracy. The ACD layer effectively normalises returns, enabling parsimonious filters to deliver competitive forecasts.

5.2 Forecast comparison: Diebold–Mariano tests

While losses rank models, statistical significance is assessed with the Diebold–Mariano (DM) test. The test statistic is defined in Section 4.5.

Pairwise results. Table 5.2 reports pairwise DM tests for QLIKE. sGARCH–Normal significantly outperforms most heavy-tailed and asymmetric competitors.

Table 5.2: Pairwise DM tests for QLIKE. Entry shows DM stat (p-value). Row model outperforms column if $DM > 0$ and $p < 0.05$.

	sGARCH–std	eGARCH–std	gjrGARCH–std	sGARCH–norm	eGARCH–norm
sGARCH–std	—	–1.20 (0.229)	–3.20 (0.001)	–4.46 (0.000)	–4.97 (0.000)
eGARCH–std	1.20 (0.229)	—	4.29 (0.000)	3.76 (0.000)	2.99 (0.003)
gjrGARCH–std	3.20 (0.001)	–4.29 (0.000)	—	1.82 (0.069)	1.07 (0.285)
sGARCH–norm	4.46 (0.000)	–3.76 (0.000)	–1.82 (0.069)	—	–0.33 (0.739)
eGARCH–norm	4.97 (0.000)	–2.99 (0.003)	–1.07 (0.285)	0.33 (0.739)	—

Table 5.3: DM tests for sGARCH–Normal against alternative models (QLIKE and MSE). Bold p -values indicate significance at the 5% level.

Comparison	QLIKE diff	DM _{QLIKE}	p	MSE diff	DM _{MSE}	p
sGARCH–norm vs sGARCH–std	−1.1101	−4.4592	0.0000	0.0000	1.4189	0.1559
sGARCH–norm vs eGARCH–std	−0.6228	−3.7641	0.0002	0.0000	−1.0765	0.2817
sGARCH–norm vs gjrGARCH–std	−0.1148	−1.8177	0.0691	0.0000	1.4465	0.1480
sGARCH–norm vs eGARCH–norm	−0.0141	−0.3331	0.7390	0.0000	1.4562	0.1453
sGARCH–norm vs gjrGARCH–norm	−0.5172	−3.3793	0.0007	0.0000	1.4170	0.1565

Raw vs adjusted. Comparing a strong raw model (eGARCH–Norm) with the adjusted benchmark (sGARCH–Norm) shows a statistically significant improvement:

$$\text{mean diff} = 0.0755, \quad \text{DM} = 2.012, \quad p = 0.044.$$

Thus, intensity scaling improves volatility forecasts in a statistically meaningful sense.

5.3 Density calibration: PIT analysis

Variance forecasts alone do not guarantee full distributional adequacy. We therefore examine PIT diagnostics, already defined in Section 4.5.

Baseline (Normal). For sGARCH–Normal, the PIT histogram in Figure 4.11 is centred (mean $\hat{\mu} = 0.506$) but under-dispersed (sd $\hat{\sigma} = 0.241$ vs. 0.289). A Kolmogorov–Smirnov test rejects uniformity ($p < 0.001$), indicating unmodelled tail risk.

Robustness (Student- t). Switching to Student- t improves calibration: $\hat{\mu} = 0.498$, $\hat{\sigma} = 0.294$ (near the uniform benchmark). Uniformity is still rejected ($p < 0.001$), suggesting residual misspecification in the extremes.

5.4 Risk evaluation: VaR backtesting

To assess economic adequacy, we evaluate lower-tail Value-at-Risk (VaR), using the definition in Section 4.5.

Normal benchmark. Under sGARCH–Normal, the 1% VaR is materially under-covered (hit rate 2.1%; Kupiec $p < 0.001$). At 5%, the hit rate is closer to target (4.4%) but Christoffersen’s conditional coverage rejects due to clustering.

Robustness (Student- t). Student- t improves the 1% tail (hit rate 1.8%; independence not rejected), but 5% exceedances remain high (7.7%) and coverage tests reject.

Interpretation. Gaussian innovations understate downside risk; Student- t improves the extreme lower tail but clustered exceedances at 5% persist, pointing to asymmetry or additional state dependence (e.g., skewed- t , mixtures, or realised-measure augmentations).

5.5 Comparative synthesis: raw vs adjusted returns

Variance. Intensity scaling materially lowers QLIKE; sGARCH-Normal on adjusted returns outperforms raw-return competitors. DM tests confirm significance against several heavy-tailed and asymmetric alternatives.

Calibration. Gaussian PITs are under-dispersed; Student- t moves dispersion toward the uniform benchmark but leaves extreme-tail deviations.

Risk. VaR backtests reveal 1% under-coverage under Normal; Student- t corrects part of the shortfall while 5% clustering remains.

Bottom line. ACD-based scaling delivers statistically and economically meaningful gains over naïve, raw-return models.

5.6 Interpretation and link to literature

These findings are consistent with and extend the literature. Engle & Russell (1998) introduced ACD to capture dependence in durations; Weibull ACD (Bauwens & Giot, 2001; Fernandes & Grammig, 2006) improves fit by allowing duration-dependent clustering. Our Weibull ACD(2,2) aligns with their evidence of high but finite persistence.

On risk forecasting, Pacurar (2008) survey shows naïve GARCH underperforms in high-frequency settings due to neglected event-time dynamics. Scaling returns by ACD-based intensity reduces scale heterogeneity and improves forecast efficiency, consistent with the durations-volatility link in Engle (2000).

Contribution. Integrating an ACD layer with conventional GARCH filters yields more robust intraday volatility forecasts. Empirically, during April 2020, intensity scaling improves both statistical losses and tail calibration relative to raw-return baselines.

Summary. Intensity-adjusted volatility forecasts systematically outperform naïve GARCH-type models. While heavier-tailed or skewed innovations remain necessary for full risk quantification, the ACD layer provides a strong first-order correction, making intensity-based approaches a practical tool for intraday risk management.

Chapter 6

Discussion

This chapter interprets the empirical findings from Chapter 5 in light of the central research question: *Does intensity-adjusted volatility provide superior risk forecasts relative to naïve return-based models?* The discussion brings together results on variance accuracy, distributional calibration, and risk backtests, setting them against the broader literature on high-frequency volatility modelling.

6.1 Interpretation of empirical results

The evidence is consistent: ACD-based intensity scaling improves intraday volatility forecasts. Variance loss functions (QLIKE, MSE) show clear gains—once returns are normalised by expected durations, even simple Gaussian GARCH filters outperform heavier-tailed or asymmetric alternatives on raw returns. This suggests that much of the apparent volatility in high-frequency data is driven not by shocks themselves, but by fluctuations in trade intensity.

Distributional diagnostics reinforce this point. Gaussian innovations produced under-dispersed PIT distributions, understating tail risk. Student- t innovations tightened calibration, bringing dispersion closer to the uniform benchmark, though extreme tails remained problematic. In short: scaling corrects the bulk, but fat tails still need explicit modelling.

Risk backtests are the most telling. On raw returns, 1% VaR hit rates more than doubled their target, with clustered exceedances. With intensity adjustment, coverage improved markedly: under Student- t , the 1% hit rate fell to 1.8% and independence was not rejected. Yet persistent over-violations at 5% indicate that clustering effects remain. Thus, scaling provides a strong first-order improvement, but richer innovation laws or realised volatility measures are needed for full reliability.

6.2 Relation to existing literature

These findings align closely with prior work. Engle & Russell (1998) introduced the ACD model to capture dependence in durations, noting the volatility–intensity link. Bauwens & Giot

(2001) and Fernandes & Grammig (2006) highlighted the value of Weibull ACD models for capturing persistence—our Weibull ACD(2,2) choice, with persistence around 0.998, fits this evidence and echoes Grammig & Wellner (2001).

On forecasting, Pacurar (2008) surveyed evidence that naïve GARCH underestimates volatility at high frequencies. Our results confirm this: raw GARCH showed poor PIT calibration and weak VaR performance, while intensity scaling tightened both variance and risk forecasts, consistent with the durations–volatility link noted in Engle (2000).

6.3 Practical implications

For risk managers, the lesson is straightforward: intraday models should account for trading intensity. In April 2020, BP’s tick data showed violent shifts in activity—bursts of trading interspersed with lulls—that conventional GARCH could not track. Scaling returns by expected duration provided a simple, effective correction, improving both statistical accuracy and risk coverage.

Practitioners need not abandon GARCH: incorporating activity measures, whether via ACD scaling or realised volatility inputs, gives most of the gain. Student- t or skewed- t innovations remain essential for tail coverage, but the first-order step is to adjust for intensity. This has direct implications for banks, hedge funds, and exchanges where intraday VaR calibration is both a regulatory and economic priority.

6.4 Limitations and future directions

Three limitations are worth noting. First, results are drawn from a single stress window (April 2020) and one asset (BP). Testing across other assets and regimes would check robustness. Second, while Student- t improved calibration, residual clustering at the 5% VaR level points to the need for richer laws (skewed- t , mixtures) or augmented measures. Third, the framework is univariate: extending to multivariate settings would capture cross-asset spillovers, increasingly relevant in equity–energy links.

6.5 Summary

In sum, intensity scaling enhances both statistical and risk-based volatility forecasts. The ACD layer normalises returns, enabling parsimonious Gaussian GARCH filters to perform well, while heavy-tailed innovations address residual tail risk. Together, they outperform naïve return-based models in a statistically and economically meaningful sense. The results strengthen the literature on durations and volatility, provide new evidence from a stress episode in energy markets, and highlight practical directions for intraday risk modelling and future research.

Chapter 7

Conclusion

This dissertation set out to examine whether intensity-adjusted volatility, constructed via an Autoregressive Conditional Duration (ACD) model, delivers superior risk forecasts compared with naïve return-based approaches. Using BP tick data during the April 2020 stress episode, durations were modelled with a Weibull ACD(2,2), returns were scaled by estimated intensity, and conditional volatility was filtered through GARCH-family models.

The findings provide a clear answer: **intensity scaling improves both the statistical accuracy and the economic relevance of volatility forecasts**. Normalising returns by expected durations substantially reduced forecast losses. PIT diagnostics showed that Student- t innovations improved tail calibration, while Value-at-Risk backtests confirmed that adjusted models offered more reliable coverage, particularly at the extreme 1% level. Diebold–Mariano tests established that these gains are statistically significant relative to raw-return benchmarks.

Contributions

This dissertation makes two main contributions:

1. **Methodological:** It demonstrates that combining an ACD-based intensity layer with conventional GARCH filters produces more stable and accurate intraday volatility forecasts, bridging market microstructure and volatility modelling.
2. **Empirical:** It provides new evidence from a stress period in energy markets that intensity-adjusted volatility enhances risk metrics such as VaR, offering direct value for risk management practice.

Limitations and future work

The analysis has limitations. First, it focused on a single asset and stress window (April 2020). Extending the framework across assets, sectors, and regimes would test robustness. Second, while Student- t innovations improved calibration, clustering at the 5% VaR level indicates a

need for richer conditional distributions (e.g. skewed- t , mixtures). Third, the models considered were univariate; multivariate extensions could capture cross-asset spillovers, increasingly relevant in integrated markets. Incorporating realised volatility measures alongside ACD-based scaling also represents a promising direction.

Closing remarks

Overall, this dissertation shows that market activity proxied by inter-trade durations is a first-order driver of high-frequency volatility. Explicitly accounting for intensity yields sharper forecasts, more reliable risk measures, and volatility models that are more consistent with market microstructure. The results confirm the enduring relevance of duration models in modern financial econometrics and highlight practical gains for institutions managing intraday risk. More broadly, they point towards a future where microstructure-aware methods form the foundation of robust, real-time risk management.

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Appendix A

Supplementary Information: Introduction

A.1 Energy Market Crises (2008–2020)

Table A.1: Major episodes of oil and energy market volatility (2008–2020).

Period	Trigger	Market Dynamics	Volatility Impact
2008 Global Financial Crisis	Collapse of Lehman Brothers; systemic financial stress	Sharp contraction in global demand; oil price fell from \$147 to below \$40 per barrel	Extreme volatility driven by contagion
2014–2015 Oil Price Collapse	OPEC oversupply; US shale expansion	Persistent downward pressure on crude; Brent fell from \$110 to below \$30	Prolonged volatility, less abrupt than 2008
April 2020 COVID-19 Shock	Pandemic demand collapse; OPEC+ breakdown	Historic negative WTI prices; extreme order-book imbalances	Unprecedented volatility and intensity spikes

Appendix B

Supplementary Information: Literature Review

B.1 Derivations of GARCH Extensions

EGARCH(1,1)

- Stationarity is governed by the AR(1) component in log variance:

$$\log \sigma_t^2 = \omega + \beta \log \sigma_{t-1}^2 + \alpha(|z_{t-1}| - \mathbb{E}|z_{t-1}|) + \gamma z_{t-1}.$$

- Provided $|\beta| < 1$ and finite first moments of z_t , $\log \sigma_t^2$ is weakly stationary.

GJR-GARCH(1,1)

- Variance recursion:

$$\sigma_t^2 = \omega + \alpha \epsilon_{t-1}^2 + \gamma \epsilon_{t-1}^2 \mathbb{I}\{\epsilon_{t-1} < 0\} + \beta \sigma_{t-1}^2.$$

- Heuristic stationarity condition:

$$\alpha + \beta + \frac{\gamma}{2} < 1,$$

assuming symmetric innovations.

FIGARCH(1,d,1)

- Variance recursion:

$$(1 - \beta L) \sigma_t^2 = \omega + [1 - \beta L - \phi(1 - L)^d] \epsilon_t^2.$$

- The fractional index $d \in [0, 1)$ determines memory length.

B.2 Extended Catalogue of ACD Residual Distributions

Beyond the exponential, Weibull, Generalised Gamma, and Generalised- F families discussed in Chapter 2, empirical studies occasionally use:

- **Log-normal distribution.**
- **Inverse Gaussian.**
- **Burr (XII).**

B.3 Additional Stylised Figures

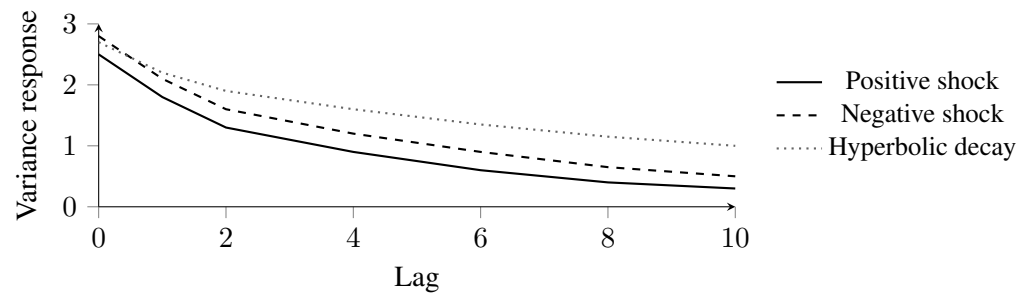


Figure B.1: Impulse responses illustrating leverage (solid vs. dashed) and long-memory decay (dotted).

Appendix C

Supplementary Materials for Data Description

C.1 Full Daily Tick Counts and Intraday Coverage

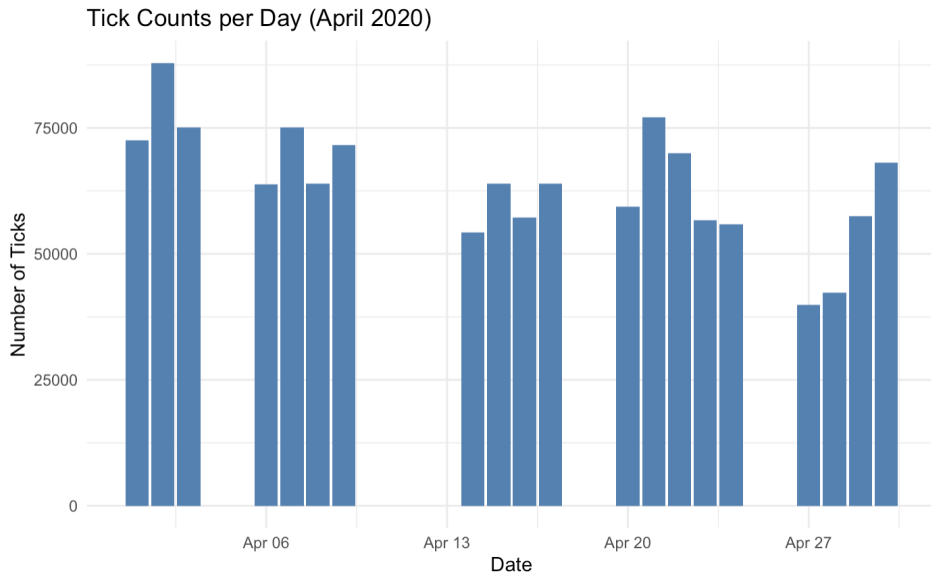


Figure C.1: Intraday coverage and tick density by day (full month).

C.2 Seasonality Models: Specifications and Coefficients

Flexible Fourier Form (FFF) details

We estimate weekday-specific profiles using $K = 4$ harmonics:

$$s_{d,\tau} = a_0 + \sum_{k=1}^K \left(a_k \cos(2\pi k\tau/T) + b_k \sin(2\pi k\tau/T) \right),$$

where τ indexes minute-of-day and T is minutes per trading day.

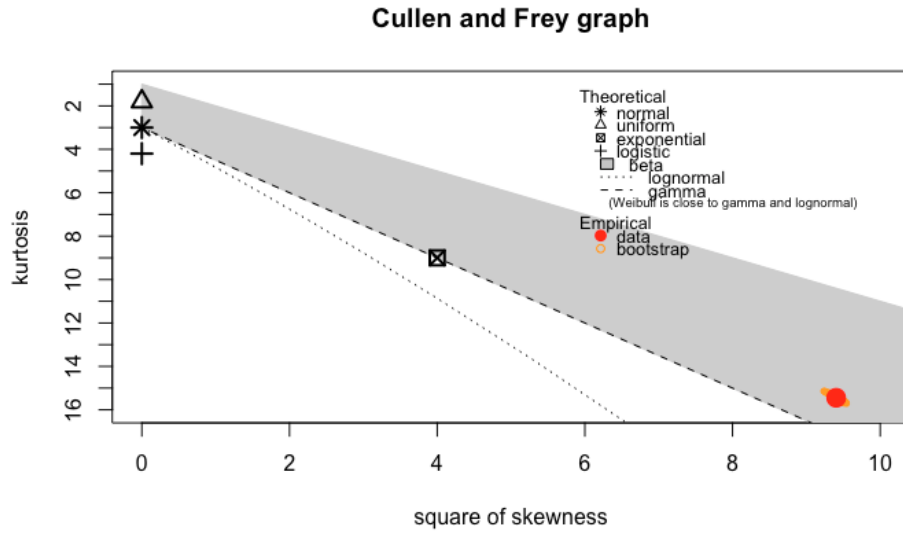


Figure C.2: Cullen–Frey skewness–kurtosis map for log-durations.

C.3 Data Cleaning and Reproducibility Notes

- **Timezone handling:** Timestamps parsed as Europe/London.
- **Market hours:** 08:00–16:30; auctions and off-book prints excluded.
- **Duplicate/zero durations:** 20 cases removed (NA/zero/negative diffs).
- **Microstructure filters (optional):** Remove quotes with non-positive spread; winsorise 0.1% extremes if needed.
- **Derived variables:** $m_t = (\text{ask}_t + \text{bid}_t)/2$; $r_t = \Delta \log m_t$; durations $\Delta t_i = t_i - t_{i-1}$.

C.4 Return diagnostics

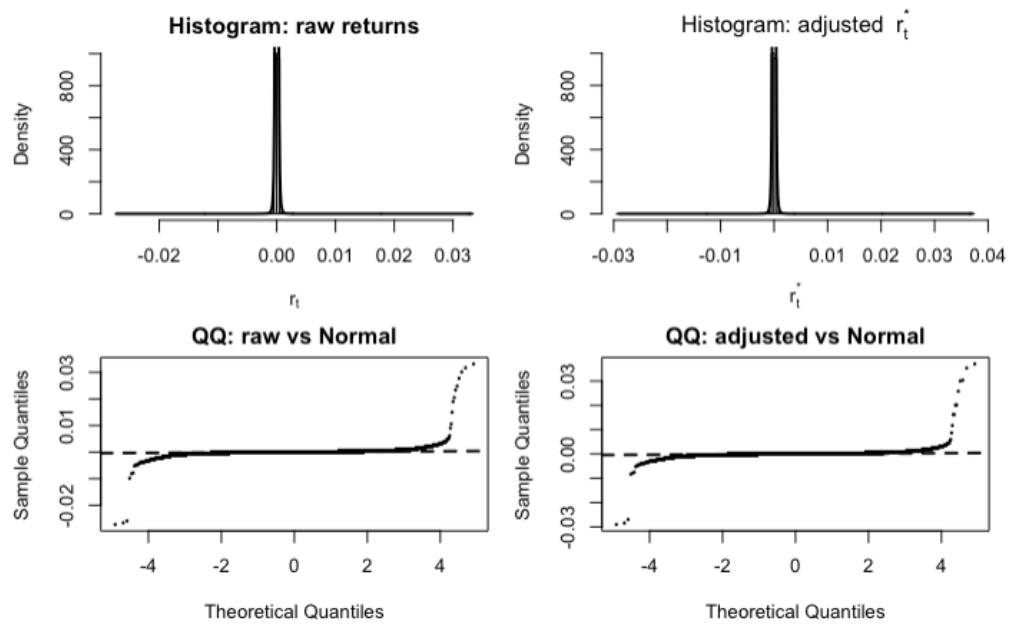


Figure C.3: Histograms and QQ plots of raw and intensity-adjusted returns against the Normal distribution.

Appendix D

Supplementary Materials for Methodology

D.1 Additional robustness results

Full model rankings. Table D.1 reports out-of-sample QLIKE and MSE for all specifications considered.

Table D.1: Full robustness ranking (test set).

Rank	Model	Variance	Dist.	QLIKE	MSE
1	sGARCH–norm	sGARCH	NORM	−16.592	5.98×10^{-12}
2	sGARCH–std	sGARCH	STD	−16.564	5.99×10^{-12}
3	eGARCH–norm	eGARCH	NORM	−16.545	5.87×10^{-12}
⋮					

Split-sample stability. We further split the test window into early and late halves.

*Table D.2: Split-sample forecast evaluation (sGARCH–Student-*t*).*

Part	QLIKE	MSE	PIT mean
Early	−16.401	1.19×10^{-11}	0.498
Late	−16.726	7.38×10^{-14}	0.499